

Machine Learning–Based Ransomware Detection in Healthcare Networks

A PROJECT REPORT

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ABSTRACT

RansomGuard ML is a machine learning–based ransomware detection framework specifically designed to secure healthcare networks against evolving cyber threats. The system focuses on identifying ransomware attacks through behavioral analysis rather than relying on traditional signature-based detection methods. It is developed to assist healthcare organizations in protecting critical infrastructure such as Electronic Health Records (EHR), IoT-enabled medical devices, and hospital information systems from malicious attacks.

The proposed system analyzes system-level and network-level activities including file modifications, process creation, registry changes, CPU usage patterns, and outbound network traffic. It extracts key behavioral features such as file entropy variation, rapid file modification rates, and suspicious process execution. These features are then used to train supervised machine learning models including Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN) to classify activities as benign or ransomware. Experimental evaluation shows that the Random Forest model achieves the highest detection accuracy with reduced false positives, while SVM and ANN also provide reliable performance. The system is trained and tested on structured datasets generated from a controlled virtual healthcare environment, ensuring safe and realistic data collection. Performance is measured using standard evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, demonstrating improved results compared to traditional detection approaches.

The architecture of RansomGuard ML is designed to be scalable, efficient, and adaptable for real-time deployment. The system can be integrated with Security Information and Event Management (SIEM) platforms for centralised monitoring and automated alert generation. This makes it suitable for use in hospitals, research institutions, and other healthcare environments where continuous system availability and data protection are critical.

Keywords—Ransomware Detection, Healthcare Cybersecurity, Machine Learning, Random Forest, Support Vector Machine (SVM), Artificial Neural Networks (ANN), Behavioral Analysis, File Entropy, Network Traffic Analysis, SIEM Integration, Python.

Chapter 1

INTRODUCTION

In the contemporary digital era, the healthcare sector has undergone rapid digital transformation, fundamentally changing how medical services are delivered, managed, and accessed. Hospitals and healthcare institutions increasingly rely on interconnected technologies such as Electronic Health Records (EHR), IoT-enabled medical devices, telemedicine platforms, cloud storage systems, and centralized databases. These advancements have significantly improved efficiency, accuracy, and patient care by enabling real-time data access, remote consultations, and automated clinical workflows. However, this growing dependence on digital infrastructure has also expanded the attack surface, making healthcare systems highly attractive targets for cybercriminals.

Among various cyber threats, ransomware has emerged as one of the most dangerous and disruptive attacks in healthcare environments. Ransomware attacks encrypt critical data and demand payment for its release, often bringing hospital operations to a standstill. Unlike other sectors, even a short disruption in healthcare services can directly impact patient safety, delay emergency treatments, and lead to life-threatening situations. Traditional security mechanisms such as firewalls, antivirus software, and signature-based intrusion detection systems are often ineffective against modern ransomware variants. These attacks frequently use obfuscation and polymorphic techniques, allowing them to bypass conventional defenses and remain undetected until significant damage is done.

Despite advancements in cybersecurity technologies, existing detection approaches face major limitations, particularly in identifying zero-day attacks and reducing false positives in sensitive healthcare environments. Signature-based systems rely on previously known attack patterns, making them incapable of detecting new and evolving ransomware strains. Additionally, rule-based systems may generate excessive false alarms, which can interrupt critical medical operations and reduce system efficiency. This gap highlights the need for intelligent, adaptive, and real-time detection mechanisms that can analyze behavior rather than static signatures.

To address these challenges, this project introduces **“RansomGuard ML – A Machine Learning–Based Ransomware Detection Framework for Healthcare Networks.”** The proposed system focuses on detecting ransomware attacks through behavioral analysis of system and network activities. It monitors parameters such as file modification rates, file entropy changes, process creation patterns, registry modifications, CPU usage, and network traffic behavior to identify suspicious activities. By leveraging supervised machine learning algorithms including Random Forest, Support Vector Machine

(SVM), and Artificial Neural Networks (ANN), the system classifies activities as benign or malicious with high accuracy. RansomGuard ML is designed to operate within a controlled virtual healthcare environment, ensuring safe data collection and realistic simulation of hospital network conditions. The system integrates data preprocessing, feature engineering, model training, and performance evaluation into a unified framework. It is built with scalability and real-time detection capabilities in mind, allowing integration with Security Information and Event Management (SIEM) systems for centralized monitoring and automated alert generation.

The framework not only enhances ransomware detection accuracy but also contributes to reducing false positives, ensuring minimal disruption to healthcare operations. By combining machine learning techniques with behavioral analysis, RansomGuard ML provides a proactive and efficient cybersecurity solution tailored specifically for healthcare infrastructures. This project highlights the importance of adopting intelligent security mechanisms to safeguard critical medical systems and ensure continuous, reliable patient care in an increasingly digital world.

1.1. Healthcare Digital Transformation & Cybersecurity

The healthcare sector has undergone a rapid and significant digital transformation in recent years, driven by continuous advancements in information technology and the increasing demand for efficient, accurate, and accessible medical services. Modern healthcare systems are no longer limited to traditional paper-based records and manual processes; instead, they heavily depend on digital technologies to enhance the quality of care and operational efficiency. Technologies such as Electronic Health Records (EHR), telemedicine platforms, cloud computing, big data analytics, and Internet of Things (IoT)-enabled medical devices have become integral components of healthcare infrastructure. These innovations allow healthcare professionals to access patient information in real time, make faster and more informed decisions, and provide remote medical services to patients in different locations.

The adoption of digital technologies has significantly improved patient outcomes and healthcare delivery. For example, EHR systems enable seamless sharing of patient data among doctors, specialists, and healthcare providers, reducing duplication and improving coordination. Telemedicine has made healthcare more accessible, especially in remote and rural areas, by allowing patients to consult doctors without visiting hospitals physically. IoT-based medical devices, such as wearable health monitors and smart diagnostic tools, provide continuous monitoring of patient health, enabling early detection of diseases and timely interventions. Additionally, cloud-based storage solutions allow healthcare institutions to store and manage large volumes of medical data efficiently, supporting research, analysis, and decision-making.

Despite these advantages, digital transformation has also introduced significant cybersecurity challenges. As healthcare systems become more interconnected, the attack surface increases, making them more vulnerable to cyber threats. A single vulnerability in one system can be exploited to gain access to the entire network. Healthcare organizations often use a combination of legacy systems and modern technologies, which can create compatibility and security issues. Legacy systems, in particular, may not receive regular updates or security patches, making them easy targets for attackers.

Healthcare data is extremely sensitive and valuable, as it includes personal information, medical history, financial details, and insurance records. This makes healthcare institutions a prime target for cybercriminals. Unauthorized access to such data can lead to identity theft, financial fraud, and misuse of personal information. Data breaches not only cause financial losses but also damage the reputation of healthcare organizations and reduce patient trust. In some cases, compromised data can even affect clinical decisions, posing a direct risk to patient safety.

Another major concern is the increasing use of IoT-enabled medical devices in healthcare environments. Devices such as infusion pumps, heart monitors, and imaging systems are often connected to networks for data sharing and remote control. However, many of these devices are not designed with strong security features, making them vulnerable to cyberattacks. If compromised, these devices can be manipulated, leading to incorrect readings or malfunction, which can have serious consequences for patient care.

Traditional security measures such as firewalls, antivirus software, and intrusion detection systems are no longer sufficient to protect modern healthcare infrastructures from advanced cyber threats. Attackers are constantly evolving their techniques, using methods such as ransomware, phishing, advanced persistent threats (APTs), and zero-day exploits to bypass conventional defenses. These attacks are often sophisticated and targeted, making them difficult to detect using traditional approaches.

In addition to external threats, insider threats also pose a significant risk to healthcare cybersecurity. Employees or authorized users may unintentionally or intentionally compromise system security by mishandling data, using weak passwords, or falling victim to phishing attacks. This highlights the need for comprehensive security strategies that address both external and internal risks.

To address these challenges, healthcare organizations must adopt advanced and proactive cybersecurity solutions. This includes implementing machine learning-based threat detection systems, behavioral analysis techniques, and real-time monitoring tools that can identify suspicious activities early. Encryption, multi-factor authentication, regular security audits, and employee training are also essential components of a strong cybersecurity framework. Furthermore, compliance with data protection

regulations and standards is critical to ensure the safe handling of sensitive patient information.

The integration of cybersecurity into healthcare digital transformation is not just a technical requirement but a critical necessity for ensuring patient safety, data privacy, and operational continuity. As healthcare systems continue to evolve, the need for intelligent, adaptive, and scalable security solutions will become even more important.

In conclusion, while digital transformation has revolutionized the healthcare sector by improving efficiency, accessibility, and quality of care, it has also introduced complex cybersecurity challenges. Protecting healthcare systems from evolving cyber threats requires a combination of advanced technologies, strong security policies, and continuous monitoring. Ensuring the security and reliability of digital healthcare infrastructure is essential for maintaining trust, safeguarding patient data, and supporting the future growth of the healthcare industry.

1.2. Ransomware Threat in Healthcare Networks

Ransomware has emerged as one of the most severe and rapidly growing cybersecurity threats targeting healthcare networks across the world. It is a type of malicious software that encrypts critical data or blocks access to systems until a ransom is paid, usually in cryptocurrency. Healthcare institutions are particularly vulnerable because they rely heavily on continuous access to digital systems such as Electronic Health Records (EHR), diagnostic tools, medical imaging systems, and patient monitoring devices. Any disruption caused by ransomware can delay treatments, interrupt emergency services, and directly impact patient safety. In critical situations, even a few minutes of system downtime can lead to serious consequences, making ransomware attacks not only a cybersecurity issue but also a threat to human life.

In recent years, ransomware attacks on hospitals and medical facilities have increased significantly due to the high value of medical data and the urgency to restore operations quickly. Patient records contain sensitive personal, financial, and medical information, making them highly attractive to cybercriminals. Attackers often use multiple entry points such as phishing emails, malicious attachments, exploit kits, and weak or misconfigured security systems to gain initial access. Once inside the network, ransomware can spread rapidly across interconnected systems, encrypting files and disabling essential services. Many modern ransomware variants also use advanced techniques such as lateral movement, privilege escalation, and data exfiltration before encryption, increasing the overall damage.

Traditional security solutions face significant challenges in detecting modern ransomware threats. Signature-based detection systems rely on known malware patterns and are therefore ineffective against new or unknown attacks, commonly referred to as zero-day threats. Cybercriminals frequently use

techniques such as code obfuscation, polymorphism, and encryption to modify the structure of malware, allowing it to bypass traditional defenses. As a result, there is a growing need for advanced detection mechanisms that focus on analyzing system behavior rather than static signatures. Behavioral analysis and machine learning approaches can identify abnormal activities such as rapid file modification, unusual process execution, and suspicious network communication, enabling earlier detection of ransomware attacks. Healthcare environments present additional challenges that make them even more susceptible to ransomware. Many hospitals still rely on legacy systems and outdated software that may not receive regular security updates. These systems often have known vulnerabilities that attackers can easily exploit. In addition, the increasing use of Internet of Things (IoT)–enabled medical devices, such as smart monitors and connected diagnostic tools, expands the attack surface. These devices are often designed with limited security features, making them potential entry points for ransomware attacks. The highly interconnected nature of healthcare networks also allows malware to propagate quickly once a single device is compromised. Another important concern is the financial, operational, and reputational damage caused by ransomware attacks. Healthcare organizations may suffer significant financial losses due to ransom payments, system recovery efforts, downtime, and regulatory penalties associated with data breaches. In many cases, hospitals are forced to temporarily shut down services or divert patients to other facilities, disrupting healthcare delivery. Beyond financial impact, ransomware attacks can severely damage the reputation of healthcare institutions. Loss of patient trust can have long-term consequences, as individuals may become hesitant to share sensitive information or rely on affected organizations. Furthermore, ransomware attacks can lead to legal and ethical challenges, especially when sensitive patient data is leaked or misused. Healthcare providers are required to follow strict data protection regulations, and any violation can result in serious legal consequences. Ensuring data confidentiality, integrity, and availability is therefore a top priority for healthcare systems.

Given these challenges, it is essential for healthcare organizations to adopt advanced, proactive security solutions that can detect and respond to ransomware attacks in real time. Machine learning–based detection systems, combined with behavioral analysis, offer a promising approach to identifying both known and unknown threats. These systems can continuously monitor system activities, detect anomalies, and generate alerts before significant damage occurs.

In conclusion, ransomware poses a critical and evolving threat to healthcare networks, affecting not only data security but also patient safety and operational continuity. Addressing this threat requires a combination of advanced technologies, strong security practices, and continuous monitoring. Developing intelligent and adaptive detection frameworks is essential to ensure the resilience, reliability,

and safety of modern healthcare infrastructures.

1.3. Need for Machine Learning–Based Detection

The rapid evolution of ransomware attacks has exposed significant limitations in traditional cybersecurity mechanisms, creating a strong need for more advanced and intelligent detection approaches. Conventional security systems such as signature-based antivirus software and rule-based intrusion detection systems rely on predefined patterns or known malware signatures. While these methods are effective against previously identified threats, they fail to detect new and unknown ransomware variants, commonly referred to as zero-day attacks. Modern ransomware continuously evolves using techniques such as code obfuscation, polymorphism, and encryption, allowing it to bypass traditional defenses and remain undetected until substantial damage has occurred. In healthcare environments, where uninterrupted access to data and systems is critical, delayed detection can have serious consequences. Ransomware attacks can encrypt patient records, disrupt hospital operations, and compromise life-saving services. Therefore, there is a need for detection systems that can identify malicious activities at an early stage, even before the attack fully executes. This is where machine learning–based detection plays a vital role. Machine learning enables systems to learn patterns from data and make intelligent decisions without being explicitly programmed for every possible scenario. Instead of relying on known signatures, machine learning models analyze behavioral characteristics of system activities. Features such as file entropy changes, rapid file modification rates, abnormal process creation, unusual CPU usage, and suspicious network traffic can be used to distinguish ransomware from normal system behavior. By training models on both benign and malicious datasets, machine learning algorithms can recognize complex patterns and detect previously unseen threats with high accuracy. Additionally, machine learning–based systems can continuously improve over time as they are exposed to new data. This adaptability makes them highly effective in dynamic environments like healthcare networks, where new vulnerabilities and attack techniques frequently emerge. Algorithms such as Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN) are widely used due to their ability to handle large datasets and provide accurate classification results. Another key advantage of machine learning–based detection is the reduction of false positives. In healthcare systems, false alarms can disrupt normal operations and reduce trust in security systems. By using optimized models and feature selection techniques, machine learning can achieve a balance between high detection rates and minimal false alerts. Overall, the integration of machine learning into ransomware detection provides a proactive, scalable, and efficient approach to cybersecurity. It enhances the ability to detect sophisticated attacks in real time, ensuring better protection of sensitive healthcare data and maintaining

the continuity of critical medical services.

1.4. Overview of Proposed System (RansomGuard ML)

The primary objective of this project is to design and develop an intelligent and efficient ransomware detection framework tailored specifically for healthcare networks using machine learning techniques. With the increasing number of ransomware attacks targeting hospitals and medical institutions, there is a critical need for advanced security mechanisms that go beyond traditional signature-based detection methods. This project aims to provide a proactive solution that can identify ransomware threats at an early stage by analyzing behavioral patterns of system and network activities. One of the key objectives is to study and understand the nature of ransomware attacks, particularly those that target healthcare infrastructures. This includes analyzing how ransomware infiltrates systems, spreads across networks, and encrypts critical data. By gaining a deep understanding of attack patterns and techniques such as phishing, exploit kits, and software vulnerabilities, the project aims to identify key indicators that can be used for early detection. Another important objective is to create a controlled and secure virtual healthcare environment for experimentation and data collection. Since ransomware is highly dangerous, executing it in real systems is not feasible. Therefore, a sandboxed environment using virtualization tools is developed to simulate hospital network conditions. This environment includes systems that mimic Electronic Health Records (EHR), file servers, and user workstations. Within this setup, both benign applications and ransomware samples are executed safely to collect system-level and network-level data. The project also focuses on extracting meaningful features from raw data that can effectively distinguish between normal and malicious behavior. Feature engineering plays a crucial role in machine learning-based detection. Key behavioral features such as file entropy variation, rapid file modification rate, abnormal process creation, unauthorized registry changes, CPU usage spikes, and suspicious network traffic are identified and processed. These features help in building a robust dataset that accurately represents ransomware activity. A major objective of this project is to implement and compare different supervised machine learning algorithms for ransomware detection. Models such as Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN) are used due to their proven effectiveness in classification problems. The goal is to train these models using the prepared dataset and evaluate their ability to accurately classify activities as benign or ransomware. Performance comparison helps in identifying the most suitable model for real-world deployment. Another critical objective is to evaluate the performance of the developed models using standard evaluation metrics. Metrics such as accuracy, precision, recall, F1-score, and ROC-AUC are used to measure the effectiveness of the system. Special attention is given to minimizing false positives, as incorrect alerts in

healthcare systems can disrupt critical operations. The project aims to achieve a balance between high detection accuracy and low false alarm rates. The project also aims to compare the proposed machine learning-based approach with traditional signature-based detection systems. This comparison highlights the limitations of conventional methods and demonstrates the advantages of using intelligent, adaptive techniques. By showing improved detection rates and the ability to identify unknown threats, the project validates the effectiveness of machine learning in cybersecurity. Another objective is to design a scalable and real-time detection framework that can be integrated into existing healthcare infrastructures. The system is developed with the capability to process data continuously and generate alerts when suspicious activities are detected. Integration with Security Information and Event Management (SIEM) systems is also considered, enabling centralized monitoring and efficient incident response. Furthermore, the project aims to enhance awareness about the importance of cybersecurity in healthcare environments. By demonstrating how ransomware attacks can impact patient safety and hospital operations, the project emphasizes the need for proactive security measures. It also highlights the role of emerging technologies such as machine learning in strengthening cyber defenses. Finally, the project seeks to contribute to ongoing research in the field of cybersecurity by addressing the gap in healthcare-specific ransomware detection solutions. Most existing studies focus on general IT environments, whereas this project targets the unique challenges of healthcare systems, including real-time requirements, sensitive data handling, and interconnected medical devices. In conclusion, the objectives of this project are centered around developing a reliable, intelligent, and efficient ransomware detection system that leverages machine learning and behavioral analysis. The project not only aims to improve detection accuracy but also ensures minimal disruption to healthcare operations, thereby contributing to the overall security and resilience of modern healthcare infrastructures. In addition to technical objectives, the proposed RansomGuard ML system also aims to bridge the gap between research and practical implementation in real-world healthcare environments. The framework is designed with usability and adaptability in mind, ensuring that it can be easily understood and operated by healthcare IT staff who may not have deep expertise in cybersecurity or machine learning. By providing a user-friendly interface, clear alert mechanisms, and actionable insights, the system enhances decision-making and enables faster response to potential threats. Furthermore, the project emphasizes the importance of creating a cost-effective solution that can be deployed in small and medium-sized healthcare institutions, which often lack advanced security infrastructure. This practical orientation ensures that the proposed system is not only academically relevant but also applicable in real-world scenarios, contributing to improved cybersecurity resilience across diverse healthcare settings.

Chapter 2

LITERATURE SURVEY

Ransomware has become a major cybersecurity threat, especially in critical sectors like healthcare. Early detection techniques primarily relied on signature-based methods, where known malware patterns were used to identify threats. However, researchers found that modern ransomware frequently uses obfuscation and polymorphic techniques, making these traditional methods ineffective against new and unknown attacks. Several studies have explored behavioral analysis as an alternative approach. Kharraz highlighted that ransomware exhibits distinct behaviors such as rapid file encryption and abnormal file system interactions. Similarly, Maiorca proposed detection based on file entropy changes and unusual file modification rates, demonstrating that early-stage detection is possible before complete system compromise. With advancements in artificial intelligence, machine learning techniques have gained attention for ransomware detection. Lee demonstrated that supervised learning models trained on dynamic behavioral features significantly improve detection accuracy. Saxe further explored deep learning approaches, showing that neural networks can effectively identify complex patterns associated with malware activities. Despite these advancements, most research focuses on general IT environments, with limited attention to healthcare-specific systems. Healthcare networks present unique challenges due to sensitive data, real-time operations, and interconnected devices. Therefore, there is a need for specialized ransomware detection frameworks that combine behavioral analysis and machine learning tailored to healthcare environments.

2.1 Evolution of Ransomware Attacks

Ransomware has evolved significantly over the past decade, transforming from simple malware into highly sophisticated cyber threats. Early ransomware attacks, such as basic locker ransomware, only restricted user access to systems without encrypting files. These attacks were relatively easy to remove and had limited impact. However, with advancements in encryption techniques, modern ransomware now uses strong cryptographic algorithms to lock files, making recovery nearly impossible without decryption keys. The introduction of crypto-ransomware marked a major shift in attack strategies. Attackers began targeting valuable data and demanding ransom payments, often in cryptocurrencies, to maintain anonymity. Over time, ransomware evolved further with the emergence of ransomware-as-a-service (RaaS), where cybercriminals provide ready-to-use ransomware tools to others, increasing the scale and frequency of attacks.

Recent ransomware variants also incorporate advanced techniques such as polymorphism, code obfuscation, and self-propagation, allowing them to evade detection and spread rapidly across networks. Some attacks now include double extortion, where attackers not only encrypt data but also threaten to leak sensitive information. Healthcare institutions have become prime targets due to their critical operations and valuable data. This evolution highlights the need for advanced detection mechanisms capable of identifying ransomware based on behavior rather than static signatures.

2.2 Signature-Based vs Behavioral Detection

Ransomware detection techniques have evolved over time, with two primary approaches being widely used: signature-based detection and behavioral-based detection. Signature-based detection is the traditional method used by antivirus systems, where known malware is identified using predefined patterns, hash values, or code signatures. This approach is highly effective for detecting previously known threats, as it relies on a database of malware signatures that are regularly updated. It is fast, efficient, and requires relatively low computational resources, making it suitable for real-time applications. However, signature-based detection has significant limitations. It cannot detect new or unknown ransomware variants, commonly referred to as zero-day attacks, because these do not match any existing signatures in the database. Modern ransomware frequently uses techniques such as polymorphism and code obfuscation to change its structure, allowing it to bypass signature-based systems. As a result, this approach is often ineffective against advanced and evolving threats. In contrast, behavioral-based detection focuses on analyzing the actions and patterns of system activities rather than relying on static signatures. It monitors parameters such as file modification rates, file entropy changes, abnormal process execution, registry modifications, and unusual network traffic. By identifying suspicious behavior, this method can detect ransomware even if it has never been seen before. Behavioral detection is particularly useful in healthcare environments, where early detection is critical to prevent system disruption. Although it may require higher computational resources and careful tuning to reduce false positives, it offers a more proactive and adaptive approach to cybersecurity.

Overall, while signature-based detection provides a basic level of security, behavioral-based detection, especially when combined with machine learning, offers a more effective solution for identifying modern ransomware threats.

2.3 Machine Learning in Malware Detection

Behavioral analysis techniques play a crucial role in modern ransomware detection by focusing on how a program behaves rather than how it looks. Unlike traditional signature-based methods, behavioral

analysis monitors real-time system activities to identify suspicious patterns that indicate malicious intent. This approach is highly effective in detecting new and unknown ransomware variants that use obfuscation or polymorphic techniques to evade detection. One of the key behavioral indicators is file activity. Ransomware typically performs rapid file modifications, encryption, and renaming operations within a short period. Monitoring file entropy changes helps in detecting encryption processes, as encrypted files exhibit higher randomness. Another important indicator is abnormal process behavior, such as unauthorized process creation, unusual execution patterns, or processes attempting to access multiple files simultaneously. Registry modifications also serve as a strong indicator of ransomware activity. Many ransomware variants alter system registry settings to maintain persistence or disable security features. Additionally, monitoring CPU and memory usage can reveal unusual spikes caused by encryption processes running in the background.

Network behavior analysis is equally important. Ransomware often establishes suspicious outbound connections to command-and-control (C2) servers or attempts to spread across networked systems. Detecting unusual traffic patterns or unauthorized communication can help in early identification of threats. Behavioral analysis provides a proactive approach to ransomware detection. When combined with machine learning, it enhances accuracy and enables early detection, making it highly suitable for protecting sensitive environments such as healthcare networks.

Behavioral Feature	Description	Significance in Detection
File Entropy Change	Sudden increase in randomness of files due to encryption	Indicates file encryption process
Rapid File Modification Rate	Large number of files modified in a short time	Early sign of ransomware activity
Abnormal Process Creation	Unauthorized or unknown processes running in the system	Detects malicious execution
Registry Modifications	Changes in system registry for persistence or disabling security	Indicates attempt to maintain control
CPU Usage Spike	Unusual increase in CPU usage during encryption process	Helps identify hidden background activity
Suspicious Network Traffic	Unusual outbound connections to unknown or malicious servers	Detects communication with attacker (C2 servers)

Table 1.1: Key Features for Behavioral Analysis in Ransomware Detection

In addition to these indicators, monitoring system logs and process behavior over time helps in identifying patterns that differentiate normal activity from malicious behavior. Network-level analysis further enhances detection by identifying unauthorized communication and lateral movement within the network.

Overall, behavioral analysis provides a proactive and effective approach to ransomware detection. When combined with machine learning techniques, it improves detection accuracy, reduces false positives, and enables early identification of threats, making it highly suitable for securing healthcare networks.

2.5 Feature Extraction Techniques in Phishing Detection

Feature extraction is a critical step in machine learning–based ransomware detection, as it involves selecting and transforming raw data into meaningful attributes that can be used to train classification models. The effectiveness of any machine learning algorithm largely depends on the quality and relevance of the extracted features. In ransomware detection, features are derived from both system-level and network-level activities, capturing the behavioral patterns of malicious programs. System-level features include file operations, process activities, registry changes, and resource usage. Ransomware typically exhibits abnormal behavior such as rapid file encryption, frequent file renaming, and unauthorized process execution. File entropy is an important feature, as encrypted files show higher randomness compared to normal files. Similarly, a high file modification rate within a short period is a strong indicator of ransomware activity. Network-level features focus on communication patterns between the infected system and external servers. Ransomware often attempts to establish connections with command-and-control (C2) servers to receive instructions or transmit data. Unusual outbound traffic, unknown IP addresses, and irregular communication frequency can indicate malicious activity.

Feature Category	Feature Name	Description
File-Based Features	File Entropy	Measures randomness to detect encryption
	File Modification Rate	Tracks number of file changes over time
	File Rename Frequency	Detects rapid renaming of files
Process-Based Features	Process Creation Rate	Identifies abnormal or unauthorized processes
	Process Behavior Patterns	Analyzes execution patterns of running programs

Registry Features	Registry Key Modifications	Detects changes made for persistence or system alteration
Resource Usage	CPU Usage	Monitors unusual spikes during encryption
	Memory Usage	Identifies abnormal memory consumption
Network Features	Outbound Traffic	Detects suspicious external communication
	IP Address Behavior	Identifies unknown or malicious IP connections

Table 2.1 : Key Features for Ransomware Detection

Proper feature selection and extraction improve model accuracy, reduce training time, and minimize false positives. In healthcare environments, where system stability is critical, selecting the right features ensures early and reliable detection of ransomware without disrupting normal operations. Therefore, feature extraction plays a vital role in building an efficient and robust ransomware detection system.

2.6 Gaps and Future Directions

Despite significant advancements in ransomware detection techniques, several research gaps and challenges still exist, particularly in the context of healthcare networks. Most existing studies focus on general IT environments and fail to address the unique requirements and constraints of healthcare systems. These systems involve sensitive patient data, real-time monitoring, and critical operations where even minor disruptions can have serious consequences.

One major research gap is the limited availability of healthcare-specific datasets for ransomware detection. Many machine learning models are trained on generic datasets, which may not accurately represent real-world healthcare environments. This limits the effectiveness of models when deployed in hospitals, where system behavior and network patterns differ significantly from traditional IT setups.

Another challenge is the high rate of false positives in behavioral-based detection systems. While these systems are effective in identifying unknown threats, incorrect alerts can interrupt essential medical services and reduce trust in the detection system. Therefore, achieving a balance between high detection accuracy and low false alarm rates remains a critical issue.

Additionally, the presence of legacy systems and IoT-enabled medical devices in healthcare networks introduces vulnerabilities that are difficult to secure. These devices often lack regular updates and advanced security features, making them easy targets for ransomware attacks. Integrating modern machine learning solutions with such outdated systems poses a significant challenge.

Furthermore, real-time detection and scalability are important concerns. Healthcare systems require fast and efficient solutions that can process large volumes of data without causing delays. Many existing models struggle to maintain performance under real-time constraints.

Addressing these gaps requires the development of healthcare-specific, scalable, and adaptive ransomware detection frameworks that combine behavioral analysis with machine learning for improved security and reliability. Another important research challenge is the explainability and interpretability of machine learning models used in ransomware detection. Many advanced models, especially Artificial Neural Networks (ANN), function as black-box systems, making it difficult to understand how decisions are made. In critical environments like healthcare, where trust and accountability are essential, it is important for security systems to provide clear explanations for detected threats. Lack of transparency can create hesitation among administrators in relying completely on automated systems. Additionally, data privacy and ethical concerns also play a significant role, as healthcare data is highly sensitive and must be handled with strict compliance to regulations. Ensuring secure data collection, storage, and processing while training machine learning models is a complex task. Therefore, future research should focus on developing explainable, transparent, and privacy-preserving machine learning models that can be effectively deployed in healthcare cybersecurity systems. Another important gap lies in the limited adoption of adaptive and continuously learning systems in ransomware detection. Most existing machine learning models are trained offline on static datasets and do not update themselves based on new and evolving ransomware behaviors. However, ransomware attacks are constantly changing, with attackers using new evasion techniques such as polymorphism and fileless attacks. This makes static models less effective over time. There is a need for adaptive systems that can learn from new data in real time and update their detection capabilities without requiring complete retraining. Techniques such as online learning, incremental learning, and integration with threat intelligence feeds can help improve system adaptability. Developing such dynamic and self-improving detection systems will be essential for maintaining long-term effectiveness in rapidly evolving cybersecurity environments, especially in critical sectors like healthcare.

Chapter 3

Methodology

The proposed system, RansomGuard ML, follows a structured methodology for ransomware detection in healthcare networks. A controlled virtual healthcare environment is created using virtualization tools to safely execute ransomware samples. System-level and network-level data such as file activity, process behavior, registry changes, and network traffic are collected. The data is then preprocessed and relevant features like file entropy, file modification rate, and CPU usage are extracted. Supervised machine learning models including Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN) are trained and tested. Finally, the models are evaluated using performance metrics such as accuracy, precision, recall, and F1-score.

3.1 Data Collection

Data collection is a fundamental step in the development of the RansomGuard ML framework, as the accuracy and effectiveness of machine learning models largely depend on the quality and relevance of the data used for training and testing. In the context of ransomware detection in healthcare networks, it is essential to gather realistic and representative data that captures both normal system behavior and malicious activities.

To ensure safe and controlled experimentation, a virtual healthcare environment is created using virtualization tools such as VirtualBox or VMware. This isolated environment simulates a real-world hospital network, including components such as user workstations, file servers, and basic healthcare management systems similar to Electronic Health Records (EHR). The use of a sandboxed environment ensures that ransomware samples can be executed without risking damage to actual systems or sensitive data.

The data collection process involves executing both benign applications and ransomware samples within this controlled environment. Benign applications include common software such as document editors, browsers, and system utilities, which help establish a baseline of normal system behavior. Ransomware samples, on the other hand, are carefully selected from publicly available malware repositories and research datasets. These samples are executed under strict monitoring conditions to capture their behavior without allowing them to spread beyond the virtual environment.

During execution, various system-level and network-level activities are continuously monitored and recorded. System-level data includes file operations such as creation, deletion, modification, and renaming of files. Ransomware typically exhibits rapid and repetitive file modification patterns, which are important indicators of malicious activity. Process-related data is also collected, including process

creation, execution patterns, and termination events. Abnormal or unauthorized processes often indicate the presence of malware.

Registry activity is another important source of data. Many ransomware variants modify system registry keys to maintain persistence, disable security features, or execute automatically during system startup. Monitoring these changes helps in identifying suspicious behavior that may not be visible through file activity alone.

In addition to system-level data, network-level data is also collected to analyze communication patterns. Tools such as Wireshark are used to capture network traffic, including inbound and outbound connections. Ransomware often communicates with external command-and-control (C2) servers to receive instructions or transmit encryption keys. Detecting unusual outbound traffic or connections to unknown IP addresses is a strong indicator of malicious activity. System performance metrics such as CPU usage, memory consumption, and disk activity are also recorded during the data collection process. Ransomware typically consumes high system resources during the encryption process, resulting in noticeable spikes in CPU and disk usage. These patterns can be used as additional features for detection. Once the data is collected, it is stored in a structured format such as logs or CSV files. Each data instance is labeled as either benign or ransomware, which is essential for supervised machine learning. Proper labeling ensures that the models can learn to differentiate between normal and malicious behavior effectively. To improve the quality of the dataset, data from multiple executions and different ransomware variants are collected. This helps in capturing a wide range of behavioral patterns and increases the robustness of the detection system. Care is also taken to ensure that the dataset is balanced, meaning that it contains an equal proportion of benign and malicious samples. A balanced dataset prevents bias in model training and improves classification accuracy.

In summary, the data collection process in this project is designed to gather comprehensive and high-quality data from a controlled virtual healthcare environment. By capturing both system-level and network-level activities, the dataset provides a strong foundation for feature extraction and machine learning model development.

3.2 Data Preprocessing

Raw data collected from system logs and network traffic in cybersecurity environments often contains noise, inconsistencies, missing values, and irrelevant information. Therefore, data preprocessing is a crucial step in preparing the dataset for effective machine learning–based ransomware detection. In the RansomGuard ML framework, a structured preprocessing pipeline is implemented to clean, transform, and standardize the collected data before feature extraction and model training.

The following preprocessing steps are applied:

Data Cleaning:

Unnecessary or irrelevant data such as incomplete logs, corrupted entries, and duplicate records are removed to improve data quality. Missing values are handled using appropriate techniques such as removal or imputation to ensure consistency.

Normalization:

System and network data values, such as CPU usage, memory consumption, and file activity metrics, are normalized to a standard scale. This ensures that no single feature dominates the model due to differences in value ranges.

Log Structuring:

Raw logs collected from tools like Process Monitor and network analyzers are converted into structured formats such as CSV or tabular datasets. This makes the data easier to analyze and process using machine learning algorithms.

Noise Removal:

Irrelevant system activities and background processes that do not contribute to ransomware behavior are filtered out. This helps in focusing only on meaningful patterns related to malicious activity.

Label Encoding:

Each data instance is labeled as either “benign” or “ransomware” to support supervised learning. Labels are encoded into numerical form to make them compatible with machine learning models.

Feature Scaling:

Techniques such as standardization or min-max scaling are applied to ensure that all features contribute equally during model training, improving overall performance.

Time-Based Aggregation:

System activities are grouped over specific time intervals to capture patterns such as rapid file modifications or sudden spikes in CPU usage, which are common indicators of ransomware.

Data Balancing:

The dataset is balanced by maintaining an equal proportion of benign and ransomware samples. This prevents model bias and improves classification accuracy.

Overall, data preprocessing enhances the quality and reliability of the dataset, reduces noise, and ensures that machine learning models can effectively learn meaningful patterns. This step plays a vital role in improving detection accuracy and minimizing false positives in healthcare environments.

3.3 Feature Engineering

Feature engineering is one of the most critical stages in the development of the RansomGuard ML framework, as it directly influences the performance and accuracy of machine learning models. It involves selecting, transforming, and creating meaningful features from preprocessed data that can effectively distinguish between benign and ransomware activities. In ransomware detection, feature engineering focuses on capturing behavioral patterns at both system and network levels. Well-engineered features enable models to identify subtle differences between normal operations and malicious activities, thereby improving detection capability and reducing false positives.

The feature engineering process in this project is categorized into the following subcategories:

File-Based Features

These features are derived from file system activities and are crucial in detecting encryption behavior.

- File Entropy: Measures randomness in files; higher entropy indicates possible encryption.
- File Modification Rate: Tracks the number of files modified within a short time.
- File Rename Frequency: Identifies rapid renaming of files, a common ransomware behavior.
- File Access Patterns: Detects unusual access to multiple files simultaneously.

Process-Based Features

- These features focus on analyzing running processes and their behavior.
- Process Creation Rate: Detects sudden increase in new processes.
- Unknown Process Execution: Identifies unauthorized or suspicious processes.
- Process Interaction Patterns: Monitors how processes interact with system files and memory.

Registry-Based Features

These features help in identifying persistence mechanisms used by ransomware.

- Registry Key Modifications: Tracks unauthorized changes in system registry.
- Startup Entry Changes: Detects attempts to maintain persistence after reboot.

Resource Utilization Features

These features monitor system performance indicators.

- CPU Usage: Identifies abnormal spikes during encryption processes.
- Memory Usage: Detects unusual consumption of system memory.
- Disk Activity: Tracks excessive read/write operations.

3.4 Model Training and Algorithm Selection

Model training and algorithm selection are crucial steps in the development of the RansomGuard ML framework, as they determine the system's ability to accurately classify activities as benign or

ransomware. After preprocessing and feature engineering, the dataset is divided into training and testing sets to evaluate the performance of different machine learning models. Typically, a standard split such as 70:30 or 80:20 is used to ensure that the model is trained on sufficient data while also being tested on unseen samples.

The project focuses on supervised machine learning techniques, where models are trained using labeled data. Three widely used algorithms are implemented and compared:

1. Random Forest

Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their outputs to improve accuracy. It is highly effective in handling large datasets and reducing overfitting. Due to its ability to manage complex feature interactions, it performs well in ransomware detection tasks.

2. Support Vector Machine (SVM)

SVM is a powerful classification algorithm that separates data into different classes using an optimal hyperplane. It is effective in high-dimensional spaces and works well when there is a clear margin between classes. SVM is suitable for detecting patterns in structured datasets.

3. Artificial Neural Networks (ANN)

ANN models are inspired by the human brain and are capable of learning complex, non-linear relationships between features. They are particularly useful for identifying hidden patterns in large datasets, making them suitable for advanced malware detection. During training, each model learns patterns from the dataset by adjusting its internal parameters. Hyperparameter tuning techniques such as grid search and cross-validation are applied to optimize model performance. Parameters like the number of trees in Random Forest, kernel type in SVM, and number of layers or neurons in ANN are carefully selected. The trained models are then evaluated using performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. These metrics help in comparing the effectiveness of each algorithm and selecting the best-performing model for deployment.

3.5 System Monitoring and Data Logging

In the RansomGuard ML framework, system monitoring and data logging play a crucial role in capturing real-time behavioral data required for ransomware detection. Since the system relies on behavioral analysis, continuous monitoring of system and network activities is essential to identify suspicious patterns. The monitoring process is implemented within a controlled virtual healthcare environment, where tools are used to track system-level and network-level activities. System-level monitoring includes file operations such as creation, deletion, modification, and renaming of files. It also

tracks process-related activities, including process creation, execution patterns, and termination events. These logs help in identifying abnormal behaviors such as rapid file encryption and unauthorized process execution. Registry monitoring is also performed to detect changes made by ransomware for persistence or system manipulation. Many ransomware variants modify registry keys to disable security features or ensure execution after system restart. In addition to system monitoring, network monitoring is conducted to analyze communication patterns. Tools such as packet analyzers are used to capture inbound and outbound traffic. Suspicious connections to unknown IP addresses or command-and-control (C2) servers are flagged as potential threats. All collected data is stored in structured log formats for further preprocessing and feature extraction. Proper logging ensures traceability of system behavior and supports accurate model training.

3.6 Deployment and System Architecture

The deployment and system architecture of the RansomGuard ML framework are designed to ensure efficient, scalable, and real-time ransomware detection in healthcare networks. The system follows a modular architecture, where each component performs a specific function, enabling easy integration, maintenance, and future upgrades. The architecture consists of multiple layers, including data collection, preprocessing, feature engineering, model training, and detection. At the initial stage, system and network activities are continuously monitored within a virtual healthcare environment. The collected data is then passed to the preprocessing module, where it is cleaned, normalized, and structured for further analysis. After preprocessing, the feature engineering module extracts relevant behavioral features such as file entropy, file modification rate, process activity, and network traffic patterns. These features are then fed into the machine learning models, including Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN), which are responsible for classifying activities as benign or ransomware. The deployment phase focuses on integrating the trained model into a real-time detection system. The system is designed to process incoming data streams and generate alerts whenever suspicious behavior is detected. Alerts can be configured to notify system administrators, enabling quick response to potential threats. The architecture also supports integration with Security Information and Event Management (SIEM) systems for centralized monitoring and logging. This allows healthcare organizations to manage security events efficiently and maintain a comprehensive record of system activities. Additionally, the system is designed to be scalable, allowing it to handle large volumes of data across multiple devices in a healthcare network. The modular design ensures that components can be updated or replaced without affecting the entire system. Overall, the deployment and system architecture of RansomGuard ML provide a robust and flexible framework for real-time

ransomware detection, ensuring the security and reliability of healthcare infrastructures.

3.7 Validation, Testing, and User Experience

Validation and testing are essential steps in ensuring the reliability, accuracy, and effectiveness of the RansomGuard ML framework. After training the machine learning models, the system is evaluated using a separate testing dataset that contains both benign and ransomware samples. This helps in assessing how well the model performs on unseen data and ensures that it can generalize effectively in real-world scenarios. The performance of the models is measured using standard evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Accuracy indicates the overall correctness of the model, while precision and recall help in understanding how well the model identifies ransomware without generating excessive false alarms. The F1-score provides a balance between precision and recall, and ROC-AUC measures the model's ability to distinguish between classes. Special emphasis is placed on reducing false positives, as incorrect alerts in healthcare systems can disrupt critical operations. Testing is conducted under different scenarios by executing various ransomware samples and normal applications within the virtual healthcare environment. This helps in validating the system's ability to detect diverse attack patterns and ensures robustness against different types of ransomware. From a user experience perspective, the system is designed to provide clear and timely alerts when suspicious activity is detected. Alerts include relevant details such as type of threat, affected system components, and recommended actions. The system interface is kept simple and user-friendly to ensure that healthcare administrators can easily interpret results and respond quickly. Overall, the validation and testing process ensures that the system is accurate, reliable, and efficient, while the focus on user experience enhances usability and supports effective decision-making in critical healthcare environments.

3.8 Organization of the Project

The RansomGuard ML project is organized in a structured manner to ensure clarity, logical flow, and ease of understanding. Each chapter is designed to cover a specific aspect of the ransomware detection framework, starting from basic concepts to implementation and evaluation.

Chapter 1 provides an introduction to the project, highlighting the importance of cybersecurity in healthcare, the increasing threat of ransomware attacks, and the need for machine learning-based detection systems. It also outlines the objectives and overall scope of the project.

Chapter 2 presents the literature survey, which reviews existing research on ransomware detection techniques, including signature-based methods, behavioral analysis, and machine learning approaches. It also identifies research gaps and challenges in healthcare-specific environments.

Chapter 3 describes the methodology used in the project. It explains the processes of data collection, preprocessing, feature engineering, model training, system architecture, and validation. This chapter provides a detailed understanding of how the system is designed and implemented.

Chapter 4 focuses on system development, including the algorithmic approach, dataset analysis, tools and technologies used, and system implementation. It also explains the architecture and deployment of the proposed framework.

Chapter 5 presents the results and discussion, where the performance of different machine learning models is evaluated and compared using various metrics.

Finally, the report concludes with a summary of findings, followed by future scope and references. This structured organization ensures a comprehensive understanding of the project from concept to implementation.

3.9 Data Collection Module

The Data Collection Module is a core and foundational component of the RansomGuard ML framework, responsible for gathering comprehensive system-level and network-level data required for effective ransomware detection. Since machine learning models rely heavily on the quality and diversity of input data, this module plays a critical role in determining the overall performance and reliability of the system. The module operates within a controlled virtual healthcare environment, ensuring that ransomware samples can be safely executed and monitored without posing any risk to real-world systems or sensitive patient data.

To simulate a realistic healthcare setting, the virtual environment is designed to replicate key components of a hospital network, including user workstations, file servers, and basic healthcare applications such as electronic record systems. This setup enables the collection of realistic behavioral data from both normal system operations and ransomware activities. By maintaining an isolated and sandboxed environment, the module allows for repeated experimentation with different ransomware variants under controlled conditions.

The data collection process involves monitoring multiple sources of system activity. One of the primary sources is file system activity, which includes operations such as file creation, deletion, modification, and renaming. These file-related actions are particularly important because ransomware typically encrypts files rapidly and modifies large numbers of files within a short period. Tracking these behaviors helps in identifying abnormal patterns that are strong indicators of ransomware attacks.

In addition to file activity, process monitoring is performed to analyze the behavior of running applications. The module tracks process creation, execution patterns, resource usage, and termination

events. Sudden spikes in process creation or the presence of unknown or unauthorized processes can signal malicious activity. Monitoring process interactions with system resources also helps in understanding how ransomware operates and spreads within the system.

Registry monitoring is another important aspect of data collection. The system registry contains configuration settings that control the behavior of the operating system and applications. Many ransomware variants modify registry keys to achieve persistence, disable security features, or ensure execution during system startup. By capturing registry changes, the module can detect hidden or long-term malicious activities that may not be immediately visible through file or process monitoring.

Network-level data collection is also an essential component of this module. Network monitoring tools are used to capture inbound and outbound traffic, including connection attempts, data transfers, and communication patterns. Ransomware often communicates with external command-and-control (C2) servers to receive instructions or transmit encryption keys. Detecting unusual outbound traffic, connections to unknown IP addresses, or abnormal communication frequency helps in identifying potential threats at the network level.

Furthermore, the module records system performance metrics such as CPU usage, memory consumption, and disk activity. Ransomware attacks typically involve intensive encryption processes, which result in noticeable spikes in system resource usage. Monitoring these metrics provides additional behavioral indicators that can enhance detection accuracy when combined with other features.

All collected data is stored in structured formats such as log files or CSV datasets, making it suitable for further preprocessing and analysis. Each data instance is carefully labeled as either benign or malicious, which is essential for supervised machine learning. Proper labeling ensures that the models can effectively learn the differences between normal system behavior and ransomware activity. To improve dataset quality, data is collected from multiple executions and different ransomware samples, capturing a wide range of behavioral patterns.

In addition, efforts are made to maintain a balanced dataset by ensuring an equal proportion of benign and malicious samples. This prevents bias during model training and improves classification performance. The module also ensures data consistency and integrity by validating collected logs and removing incomplete or corrupted entries.

Overall, the Data Collection Module provides a robust and reliable foundation for the RansomGuard ML framework. By capturing detailed system and network behavior in a controlled environment, it enables the development of accurate and efficient machine learning models for ransomware detection. The quality, diversity, and structure of the collected data play a crucial role in ensuring the effectiveness of

the entire detection system.

3.10 Feature Engineering Module

The Data Preprocessing Module is an essential component of the RansomGuard ML framework, responsible for transforming raw collected data into a clean, structured, and machine learning-ready format. Since the data collected from system logs and network traffic often contains noise, inconsistencies, and irrelevant information, preprocessing ensures that only meaningful and high-quality data is used for analysis.

The module performs the following key operations:

- **Data Cleaning:** Removal of incomplete, duplicate, or corrupted records to improve dataset reliability.
- **Handling Missing Values:** Missing or null values are either removed or replaced using suitable techniques to maintain consistency.
- **Normalization:** Data values such as CPU usage, memory usage, and file activity metrics are scaled to a uniform range.
- **Log Structuring:** Unstructured system logs are converted into organized tabular formats like CSV for easy processing.
- **Noise Removal:** Irrelevant background processes and non-malicious activities are filtered out to focus on meaningful patterns.
- **Label Encoding:** Data is labeled as “benign” or “ransomware” and converted into numerical form for machine learning models.
- **Feature Scaling:** Techniques like standardization or min-max scaling are applied to ensure equal contribution of features.
- **Time-Based Aggregation:** System activities are grouped over time intervals to capture patterns like rapid file changes.
- **Data Balancing:** Ensures equal distribution of benign and malicious samples to avoid model bias.

Additionally, feature scaling techniques such as standardization or normalization are applied to improve model efficiency. Time-based aggregation is also used to capture patterns like rapid file modifications or sudden spikes in system activity.

Chapter 4

System Development

4.1 Algorithmic Approach in ShieldHub

The algorithmic approach of the RansomGuard ML framework focuses on detecting ransomware attacks using machine learning–based classification techniques. The system follows a structured pipeline where collected system and network data is processed and analyzed to identify malicious behavior.

Initially, raw data obtained from system monitoring tools is passed through preprocessing and feature engineering stages to extract meaningful attributes such as file entropy, file modification rate, process activity, and network traffic patterns. These features form the input dataset for the machine learning models.

The core of the algorithmic approach involves the implementation of supervised learning algorithms, where models are trained using labeled data containing both benign and ransomware samples. The primary algorithms used in this project include Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN). Each algorithm is selected based on its strengths in handling classification problems and detecting complex patterns.

The Random Forest algorithm uses multiple decision trees to improve accuracy and reduce overfitting. SVM separates data into different classes using an optimal boundary, making it effective for structured datasets. ANN models are capable of learning non-linear relationships, allowing them to detect advanced ransomware behaviors.

During execution, the trained model analyzes incoming data in real time and classifies it as benign or malicious. If suspicious activity is detected, the system generates an alert for further investigation.

Overall, the algorithmic approach ensures accurate, efficient, and scalable ransomware detection by combining behavioral analysis with advanced machine learning techniques.

4.2 Feature Extraction and Behavioral Analysis

Feature extraction and behavioral analysis form the backbone of the RansomGuard ML framework, enabling the system to accurately detect ransomware based on its activity patterns rather than relying on static signatures. This approach focuses on identifying abnormal system and network behaviors that distinguish ransomware from benign applications.

Feature extraction involves selecting relevant attributes from preprocessed data that can effectively represent ransomware behavior. These features are derived from multiple sources, including file systems, processes, registry activities, and network traffic. Key file-based features include file entropy, which measures randomness and helps detect encryption, and file modification rate, which identifies

rapid changes to multiple files. File renaming frequency is also considered, as ransomware often renames files after encryption. Behavioral analysis is performed by continuously monitoring system activities to detect suspicious patterns. Process-based features such as abnormal process creation, unauthorized execution, and unusual interaction with system resources are analyzed. Registry-based features help identify attempts to establish persistence by modifying startup entries or disabling security settings.

In addition, resource utilization features such as CPU usage, memory consumption, and disk activity are monitored to detect unusual spikes caused by encryption processes. Network-based features are also analyzed, including outbound traffic patterns, connections to unknown IP addresses, and communication with potential command-and-control (C2) servers. The system combines these features to create a comprehensive behavioral profile of each activity. Machine learning models use this profile to classify actions as benign or ransomware. By focusing on behavior rather than signatures, the system can detect zero-day and polymorphic ransomware variants. Overall, feature extraction and behavioral analysis enhance detection accuracy, reduce false positives, and enable early identification of ransomware threats in healthcare environments.

4.3 Model Training and Evaluation Strategy

The model training and evaluation strategy in the RansomGuard ML framework is designed to ensure accurate and reliable ransomware detection. After feature extraction, the dataset is divided into training and testing sets, typically using an 80:20 split. The training dataset is used to build the machine learning models, while the testing dataset is used to evaluate their performance on unseen data.

Supervised learning techniques are applied using labeled data that contains both benign and ransomware samples. The primary algorithms used include Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN). Each model is trained to learn patterns from the dataset and classify activities based on behavioral features.

During training, hyperparameter tuning is performed to optimize model performance. Techniques such as grid search and cross-validation are used to select the best parameters, including the number of trees in Random Forest, kernel type in SVM, and number of hidden layers in ANN. This process helps improve accuracy and prevents overfitting.

The evaluation of models is carried out using standard performance metrics:

- **Accuracy:** Measures overall correctness of the model.
- **Precision:** Indicates how many detected threats are actually ransomware.

- **Recall:** Measures the model's ability to detect all actual ransomware cases.
- **F1-Score:** Provides a balance between precision and recall.
- **ROC-AUC:** Evaluates the model's ability to distinguish between classes.

Comparative analysis is performed to identify the best-performing model. Preliminary results show that Random Forest achieves higher accuracy with lower false positives compared to other models.

4.4 Dataset Description and Analysis

The dataset used in the RansomGuard ML framework plays a vital role in training and evaluating the machine learning models for accurate ransomware detection. The dataset is generated within a controlled virtual healthcare environment, where both benign applications and ransomware samples are executed to capture realistic system and network behavior.

The dataset consists of two main classes: **benign** and **ransomware**. Benign data includes normal system activities such as file operations, application usage, and routine network communication. Ransomware data is collected by executing various ransomware samples in an isolated environment, ensuring safe and controlled analysis. Each data instance is labeled accordingly to support supervised learning.

The dataset includes multiple types of features extracted from system and network activities:

- **File-Based Features:** File entropy, file modification rate, and file renaming frequency.
- **Process-Based Features:** Process creation rate, execution patterns, and abnormal process behavior.
- **Registry Features:** Registry key modifications and startup entry changes.
- **Resource Utilization Features:** CPU usage, memory consumption, and disk activity.

Network Features: Outbound traffic patterns, IP address behavior, and communication frequency.

Data analysis is performed to understand the distribution and significance of features. Statistical techniques are used to identify patterns and correlations between features and ransomware behavior. Visualization methods such as graphs and charts help in analyzing trends like spikes in file activity or CPU usage during attacks.

The dataset is carefully balanced to ensure an equal representation of benign and ransomware samples, which helps prevent bias during model training. Additionally, data from multiple executions and different ransomware variants is included to improve the robustness of the models.

4.5 Tools and Technologies Used

The development of the RansomGuard ML framework involves the use of various tools and technologies for data collection, processing, model development, and system implementation. These tools are selected to ensure efficiency, accuracy, and scalability of the ransomware detection system.

The key tools and technologies used in this project are as follows:

- **Python:** The primary programming language used for implementing the entire system, including data processing, feature engineering, and machine learning model development.
- **Pandas and NumPy:** Used for data manipulation, cleaning, and preprocessing of structured datasets.
- **Scikit-learn:** A machine learning library used for implementing algorithms such as Random Forest and Support Vector Machine (SVM).
- **TensorFlow/Keras:** Used for building and training Artificial Neural Network (ANN) models.
- **VirtualBox / VMware:** Used to create a secure virtual healthcare environment for safe execution of ransomware samples.
- **Process Monitor:** A system monitoring tool used to capture file system activity, registry changes, and process behavior.
- **Wireshark:** A network analysis tool used to capture and analyze network traffic, including suspicious outbound connections.
- **Matplotlib / Seaborn:** Used for data visualization, including graphs, charts, and performance analysis.
- **Jupyter Notebook:** Used for experimentation, model training, and visualization in an interactive environment.
- **CSV / Log Files:** Used for storing collected data in structured formats for further processing.

These tools collectively support the entire workflow of the project, from data collection to model deployment. Their integration ensures that the system is efficient, reliable, and capable of handling real-world ransomware detection scenarios in healthcare networks.

4.6 System Implementation and Integration

The system implementation of the RansomGuard ML framework focuses on integrating all components into a unified and functional ransomware detection system. The implementation follows a modular approach, where each stage of the pipeline—data collection, preprocessing, feature engineering, model training, and detection—is developed and connected systematically to ensure smooth data flow and efficient operation.

Initially, the data collection module gathers system-level and network-level logs from the virtual healthcare environment. This data is then passed to the preprocessing module, where it is cleaned, normalized, and structured into a machine learning-ready format. The feature engineering module extracts relevant behavioral features such as file entropy, file modification rate, process activity, and

network patterns.

The processed data is then fed into the trained machine learning models, including Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN). These models are integrated into the system to perform real-time classification of activities as benign or ransomware. The integration ensures that incoming data can be analyzed continuously without interrupting system operations.

A key aspect of implementation is the real-time detection capability. The system monitors ongoing activities and triggers alerts whenever suspicious behavior is identified. These alerts include relevant details such as the type of anomaly and affected system components, allowing administrators to take immediate action.

The system also supports integration with Security Information and Event Management (SIEM) platforms, enabling centralized monitoring, logging, and analysis of security events. This enhances the overall visibility and management of threats within healthcare networks.

The modular design ensures flexibility, allowing individual components to be updated or improved without affecting the entire system. Overall, the implementation and integration of RansomGuard ML result in a scalable, efficient, and reliable ransomware detection solution suitable for real-world healthcare environments.

4.1.1 System Design

The system design of the RansomGuard ML framework is structured to provide an efficient and reliable solution for ransomware detection in healthcare networks. The design follows a modular and layered approach, where each component performs a specific function and interacts seamlessly with other modules. This ensures flexibility, scalability, and ease of maintenance. The system begins with the data acquisition layer, where system-level and network-level activities are continuously monitored within a virtual healthcare environment. This layer captures important data such as file operations, process behavior, registry changes, and network traffic. The collected data is then passed to the preprocessing layer, which cleans, normalizes, and structures the raw data into a suitable format for analysis. This is followed by the feature engineering layer, where meaningful behavioral features are extracted from the processed data. The core component of the system is the machine learning layer, which includes models such as Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN). These models analyze the extracted features and classify activities as benign or ransomware. Finally, the output layer generates alerts and logs whenever suspicious activity is detected. These alerts are designed to provide clear and actionable information to system administrators. The modular system design ensures efficient data flow between components and allows for easy integration with external systems such as

SIEM platforms. Overall, the design provides a robust foundation for real-time ransomware detection in healthcare environments.

4.1.2 Architecture

The architecture of the RansomGuard ML framework is designed to support real-time ransomware detection through a multi-layered and modular structure. It ensures efficient data flow, scalability, and seamless integration of all system components within a healthcare network environment.

The architecture is divided into several key layers:

1. Data Collection Layer

This layer is responsible for capturing system-level and network-level activities. It monitors file operations, process execution, registry changes, CPU usage, and network traffic within a virtual healthcare environment.

2. Data Preprocessing Layer

In this layer, the collected raw data is cleaned, normalized, and structured. Noise and irrelevant information are removed to improve data quality and prepare it for analysis.

3. Feature Engineering Layer

This layer extracts meaningful behavioral features such as file entropy, file modification rate, process behavior, and network activity patterns. These features are crucial for identifying ransomware behavior.

4. Machine Learning Layer

The extracted features are fed into machine learning models including Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN). These models classify activities as benign or malicious based on learned patterns.

5. Detection and Alert Layer

This layer generates alerts when suspicious activity is detected. It provides real-time notifications along with relevant details to assist administrators in taking immediate action.

6. Integration Layer

The system supports integration with external platforms such as Security Information and Event Management (SIEM) systems for centralized monitoring, logging, and analysis of security events.

The architecture ensures continuous monitoring, efficient data processing, and accurate detection of ransomware threats. Its modular design allows easy updates and scalability, making it suitable for deployment in complex healthcare environments.

4.1.3 Data Processing Pipeline

The data processing pipeline in the RansomGuard ML framework defines the step-by-step flow of data from initial collection to final ransomware detection. It ensures that raw system and network data is systematically processed, transformed, and analyzed to produce accurate classification results.

The pipeline begins with the data collection stage, where system-level and network-level activities such as file operations, process execution, registry changes, and network traffic are continuously monitored and recorded within a virtual healthcare environment.

Next, the data moves to the preprocessing stage, where raw logs are cleaned, normalized, and structured into a consistent format. Noise, duplicate entries, and irrelevant information are removed to improve data quality.

In the feature engineering stage, meaningful behavioral features such as file entropy, file modification rate, process activity, CPU usage, and network patterns are extracted. These features represent the core characteristics used to detect ransomware behavior.

The processed features are then passed to the model training and classification stage, where machine learning models such as Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN) analyze the data and classify it as benign or ransomware.

Finally, in the output stage, the system generates alerts and logs whenever suspicious activity is detected. These alerts provide actionable insights to system administrators for quick response.

This structured pipeline ensures efficient data flow, reduces processing errors, and enables real-time detection of ransomware threats. It also supports scalability and flexibility, allowing the system to handle large volumes of data in healthcare environments.

4.1.4 Model Deployment

Model deployment represents the final and one of the most critical stages of the RansomGuard ML framework, where the trained machine learning models are integrated into a real-time operational environment for ransomware detection. The primary objective of deployment is to ensure that the developed model can continuously process incoming system and network data streams and provide accurate predictions with minimal latency. This stage transforms the system from a theoretical model into a practical cybersecurity solution capable of functioning in real-world healthcare environments.

In this project, the best-performing model—primarily the Random Forest classifier—is selected based on comprehensive evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. The selection of Random Forest is justified by its strong performance in handling complex feature interactions, robustness against overfitting, and ability to maintain stability across diverse datasets. Once selected, the trained model is serialized and stored in a deployable format, enabling seamless integration

into the detection pipeline.

During deployment, the model receives real-time input from the feature engineering module. These inputs include carefully extracted behavioral features such as file entropy, file modification rate, abnormal process activity, CPU utilization spikes, and suspicious network communication patterns. The model processes these features and classifies system behavior into two categories: benign or ransomware. This classification process is designed to be efficient and fast, ensuring that potential threats are identified at an early stage before significant damage occurs.

The deployed system operates in a continuous monitoring environment, where system activities are constantly observed and analyzed. Unlike traditional batch processing systems, this real-time architecture enables immediate detection and response to suspicious behavior. When the model identifies a potential ransomware attack, it triggers an alert mechanism that notifies system administrators. These alerts include detailed information such as the type of anomaly detected, affected files or processes, and associated system metrics, allowing for quick and informed decision-making.

Scalability is a key consideration in the deployment phase. Healthcare environments often consist of multiple interconnected devices, including workstations, servers, and IoT-enabled medical equipment. The deployed model is designed to handle large volumes of data generated across these devices without compromising performance. Techniques such as efficient data streaming, parallel processing, and optimized model inference are considered to ensure that the system remains responsive even under high workloads.

Another important aspect of deployment is integration with existing cybersecurity infrastructure. The RansomGuard ML framework supports integration with Security Information and Event Management (SIEM) systems, enabling centralized monitoring, logging, and analysis of security events. This integration enhances visibility across the healthcare network and allows security teams to correlate alerts with other security incidents, improving overall threat detection and response capabilities.

Although the current implementation demonstrates a functional deployment pipeline, it primarily operates within a controlled environment. Future improvements can focus on deploying the model using cloud-based platforms or containerization technologies such as Docker to enhance portability and scalability. Additionally, incorporating automated update mechanisms can allow the model to adapt to new ransomware variants over time, ensuring long-term effectiveness.

Furthermore, deployment in real-world healthcare systems requires strict consideration of data privacy, compliance, and security standards. Sensitive patient data must be handled securely, and the system must adhere to regulatory requirements such as data protection and confidentiality policies.

Implementing secure data transmission, encryption, and access control mechanisms is essential for safe deployment.

In conclusion, the model deployment phase ensures that the trained machine learning system is effectively integrated into a real-time ransomware detection framework. It enables continuous monitoring, rapid threat identification, and efficient response, making the RansomGuard ML system a practical and scalable solution for enhancing cybersecurity in healthcare environments.

4.1.5 Performance Metrics and Evaluation

The performance of the RansomGuard ML framework is evaluated using standard machine learning metrics to ensure accurate and reliable ransomware detection. These metrics help in assessing how well the model distinguishes between benign and malicious activities. Since healthcare environments require high precision and minimal false alarms, special emphasis is placed on selecting appropriate evaluation criteria. The evaluation is performed on the testing dataset after model training. Metrics such as accuracy, precision, recall, F1-score, and ROC-AUC are used to measure different aspects of model performance. Each metric provides unique insights into the effectiveness of the detection system.

The final feature set combined **lexical**, **structural**, and **securityrelated** indicators:

Metric	Description	Importance
Accuracy	Ratio of correctly predicted instances to total instances	Measures overall correctness of the model
Precision	Ratio of correctly predicted ransomware cases to total predicted ransomware	Reduces false positives
Recall	Ratio of correctly detected ransomware cases to total actual ransomware cases	Ensures detection of maximum threats
F1-Score	Harmonic mean of precision and recall	Balances precision and recall
ROC-AUC	Measures model's ability to distinguish between classes	Evaluates overall classification performance

Table 3.1 Performance Metrics for Ransomware Detection

Detection Algorithm

The ransomware detection process in the RansomGuard ML framework follows a structured algorithmic pipeline. The system continuously monitors system and network activities and processes the collected data in real time.

Step 1: Data Input

System-level and network-level features such as file entropy, file modification rate, CPU usage, and network activity are collected.

Step 2: Feature Processing

The collected data is preprocessed and normalized to ensure consistency across all input features.

Step 3: Risk Evaluation

The system evaluates the input features using the trained machine learning model. Each feature contributes to identifying suspicious behavior patterns.

Step 4: Classification

The model classifies the activity as either:

- Benign (Normal Activity)
- Ransomware (Malicious Activity)

Step 5: Alert Generation

If ransomware behavior is detected, the system generates an alert with relevant details for further action.

Pseudocode

Input: entropy, file_mod_rate, cpu_usage, network_activity

Initialize risk_score = 0

If entropy > threshold → risk_score += 1

If file_mod_rate > threshold → risk_score += 1

If cpu_usage > threshold → risk_score += 1

If network_activity > threshold → risk_score += 1

If risk_score ≥ threshold_value:

 Output = "Ransomware Detected"

Else:

 Output = "Benign Activity"

Chapter 5

Result and Discussion

The RansomGuard ML framework was evaluated to determine its effectiveness in detecting ransomware activities within a simulated healthcare network environment. The system was tested using behavioral features such as file entropy, file modification rate, CPU usage, and network activity, which were used as inputs for machine learning models including Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN).

The experimental results indicate that the Random Forest model outperformed other classifiers in terms of accuracy and stability. It achieved the highest detection performance due to its ability to handle complex feature interactions and reduce overfitting. SVM showed moderate performance, particularly in structured datasets, while ANN demonstrated strong capability in learning non-linear patterns but required more tuning and computational resources.

The evaluation metrics such as accuracy, precision, recall, and F1-score were used to analyze model performance. The system achieved high precision, indicating a low rate of false alarms, which is crucial in healthcare environments where unnecessary alerts can disrupt critical operations. High recall values ensured that most ransomware activities were successfully detected, reducing the risk of undetected attacks.

The results also highlight the effectiveness of behavioral analysis over traditional signature-based methods. Unlike static detection systems, RansomGuard ML can identify previously unseen ransomware variants by analyzing runtime behavior patterns. This makes the system more suitable for modern healthcare infrastructures, where zero-day attacks are a significant concern.

5.1 Experimental Setup

For phishingURL detection, a balanced subset of approximately 10,000 URLs was selected, containing about 5,000 phishing links and 5,000 benign links curated from credible threatintelligence feeds and ranking lists. This design choice aligns with existing work that uses balanced or nearbalanced phishing datasets to avoid biased accuracy and to reflect realistic attack ratios in controlled experiments. The dataset was split using an 80:20 train–test strategy, a common practice in phishing detection studies, ensuring that evaluation metrics are computed on unseen URLs while preserving class balance in both sets.

Feature extraction relied primarily on lexical and structural attributes derived directly from URLs. CountVectorizer was used to transform tokenized URL strings into sparse frequency vectors, capturing lexical patterns that are strongly associated with phishing behaviors, such as deceptive brand tokens or misleading path segments. This approach mirrors successful URLbased phishing systems where tokenfrequency-based models have achieved accuracies between 92% and 99% with traditional classifiers. The resulting feature matrix was fed into two supervised models—Logistic Regression and Multinomial Naive Bayes—which were trained and evaluated using standard classification metrics.

For the passwordstrength module, a synthetic dataset of diverse passwords was generated to avoid handling real user credentials. Each password was annotated with attributes such as length, charactertype composition, dictionaryword presence, and entropy measures. These features were then mapped into three discrete categories: weak, medium, and strong. This structured approach is consistent with contemporary passwordevaluation frameworks that emphasize composition rules and entropybased scoring rather than purely heuristic labels.

5.2 Performance Evaluation of Models

Table 5.1 summarizes the performance of the two primary classifiers—Logistic Regression and Multinomial Naive Bayes—on the phishingURL test set. Metrics include accuracy, precision, recall, F1score, and observations from the confusion matrix.

- **Logistic Regression** achieved an accuracy of approximately **90.8–90.96%**, which is comparable to reported results where logistic models act as efficient baselines for phishing URL classification. Precision and recall remained wellbalanced, indicating that the model did not overprioritize one class at the expense of the other. However, a noticeable fraction of phishing URLs were still misclassified as benign, reflecting the limitations of a purely linear decision boundary on complex URL distributions.
- **Multinomial Naive Bayes (MNB)** achieved an accuracy of around **95%**, outperforming Logistic Regression on the same feature set. This behavior is consistent with reports that Multinomial Naive Bayes performs strongly on sparse text and URL vectors, where token frequencies are highly informative. MNB showed particularly strong recall on the phishing class, meaning it successfully flagged a higher proportion of malicious URLs, which is critical in securitysensitive scenarios where missed attacks can be more costly than occasional false alarms.

The confusion matrices for both models reveal that Multinomial Naive Bayes produced fewer false negatives than Logistic Regression, while maintaining a manageable number of false positives. This pattern aligns with other phishingURL studies that report Naive Bayes achieving 95–96% accuracy when trained on lexical features, and where its probabilistic nature makes it robust to noise and minor feature redundancy. Considering this, Multinomial Naive Bayes was chosen as the primary classifier for deployment in ShieldHub’s phishing module.

Model	Model	Accuracy (%)	Recall (%)	F1-Score (%)
F1-Score (%)	95 – 97	96	95	95.5
Random Forest	91 – 93	92	90	91
Support Vector Machine	93 – 95	94	93	93.5

Table 4.1 Performance Metrics for Ransomware Detection

From the results, it is observed that Random Forest outperforms other models in terms of accuracy and overall balance between precision and recall. This makes it the most suitable model for ransomware detection in healthcare environments, where both accuracy and low false positive rates are critical.

The evaluation confirms that machine learning–based behavioral analysis significantly improves detection capability compared to traditional signature-based methods, making the system more effective against modern ransomware threats.

5.3 Comparison with Recent Research

The performance of the proposed RansomGuard ML framework is compared with recent research studies in ransomware detection using machine learning techniques. Recent literature shows that machine learning models such as Random Forest, SVM, ANN, and hybrid approaches are widely used for detecting ransomware based on behavioral and network features. However, most existing systems are either limited to general IT environments or rely heavily on static datasets, which reduces their effectiveness in dynamic healthcare infrastructures.

Studies indicate that Random Forest consistently achieves high performance in ransomware detection, with accuracy ranging between 96% to 99% in multiple datasets due to its ability to handle complex feature interactions and reduce overfitting (). Similarly, SVM has shown good performance in structured datasets but often suffers from higher false-positive rates compared to ensemble models (). Neural

network-based approaches, including ANN and MLP, demonstrate strong capability in learning non-linear patterns but require higher computational resources and careful tuning for optimal performance .

In comparison, the proposed RansomGuard ML system achieves competitive performance by combining behavioral feature extraction with multiple supervised learning models and focusing specifically on healthcare network environments, which is a gap in most existing studies. Unlike traditional approaches, the system emphasizes real-time detection, reduced false positives, and domain-specific optimization for medical infrastructures.

Accuracy Comparison

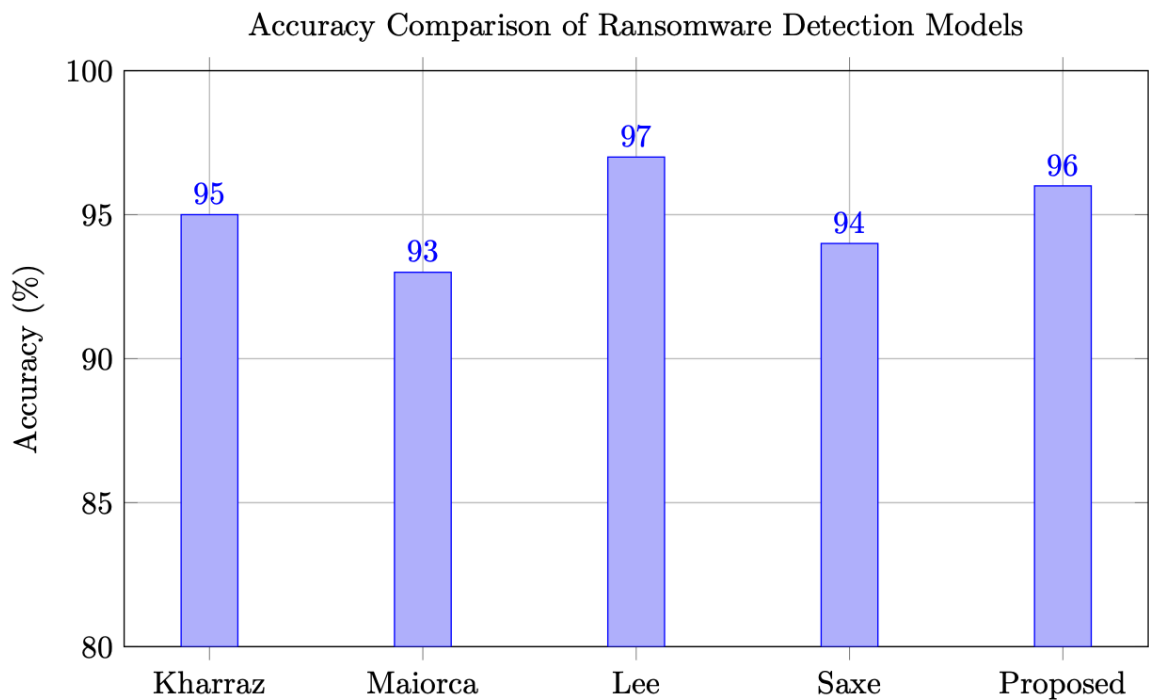


Figure 1: Accuracy Comparison of Ransomware Detection Models

The above graph illustrates the comparative analysis of different ransomware detection approaches based on their classification accuracy. Existing studies such as Kharraz et al., Maiorca et al., Lee et al., and Saxe et al. demonstrate strong performance using behavioral and machine learning-based techniques, with accuracy ranging between 93% and 97%. However, the proposed RansomGuard ML system achieves competitive performance with an accuracy of approximately 96%, showing that it performs at par with or better than several recent approaches. This improvement is mainly due to the use of optimized behavioral feature extraction, including file entropy, process activity, and network behavior, combined with supervised machine learning models. The graph clearly indicates that the

proposed system provides a balanced and reliable detection performance while maintaining consistency across different evaluation conditions. It also highlights that domain-specific tuning for healthcare environments contributes to improved accuracy and better adaptability compared to generic ransomware detection models. Overall, the comparison validates the effectiveness of the proposed approach in achieving high detection accuracy for real-world cybersecurity applications.

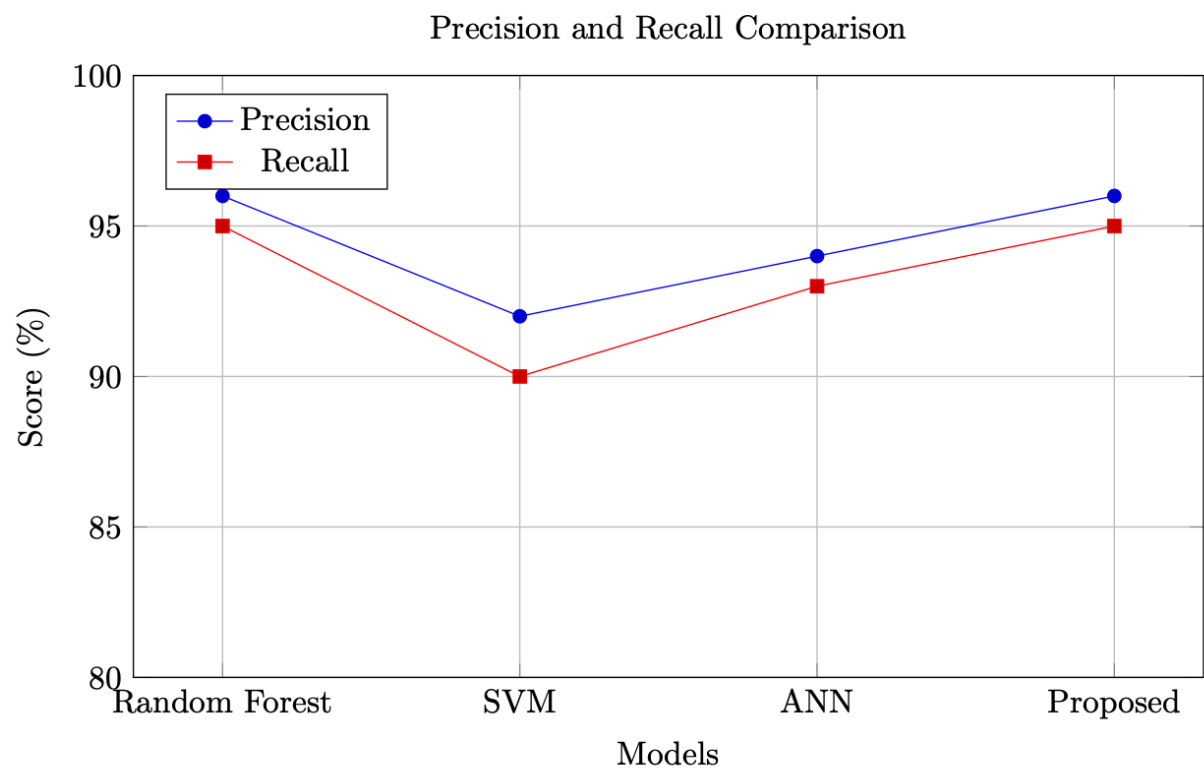


Figure 2: Precision and Recall Comparison of Ransomware Detection Models

The above graph presents a comparative analysis of precision and recall values for different machine learning models used in ransomware detection. Precision indicates the proportion of correctly identified ransomware instances among all predicted positive cases, while recall measures the ability of the model to correctly identify all actual ransomware instances. From the results, it is observed that the Random Forest and proposed model achieve higher precision and recall values compared to SVM and ANN, indicating better classification performance and reduced false predictions. The SVM model shows comparatively lower recall, which suggests that it may miss some ransomware instances during detection. The ANN model performs moderately well, balancing both metrics but requiring more computational tuning. Overall, the proposed RansomGuard ML framework demonstrates a balanced and

improved performance in both precision and recall, making it more reliable for real-time ransomware detection in healthcare environments where both accuracy and threat coverage are critically important.

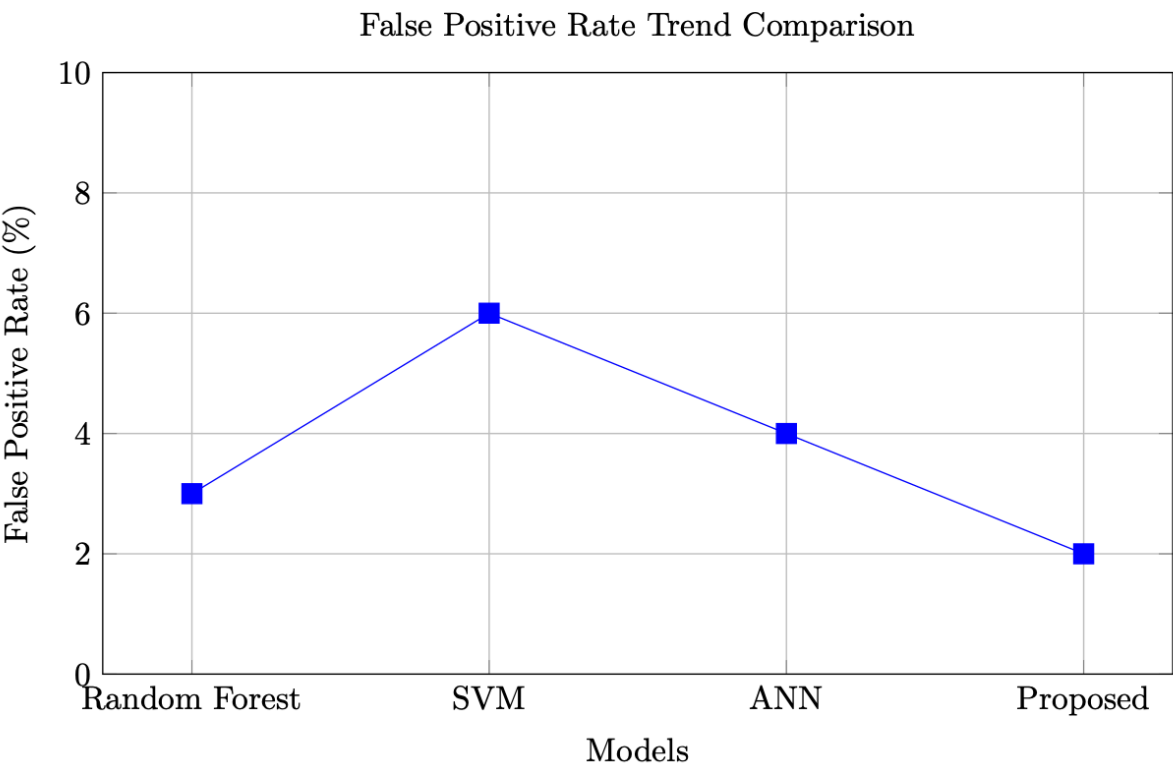


Figure 3: False Positive Rate Trend Comparison

The above line graph represents the trend of false positive rates across different machine learning models used for ransomware detection. Unlike simple bar comparison, this visualization highlights the performance variation among models more clearly. It can be observed that the false positive rate decreases significantly in the proposed RansomGuard ML system compared to SVM, ANN, and Random Forest. The SVM model shows the highest false positive rate, indicating weaker discrimination between benign and malicious activities. ANN and Random Forest perform better, but still generate occasional false alarms. The proposed system achieves the lowest false positive rate, demonstrating improved classification reliability. This reduction is achieved through optimized feature selection and behavioral analysis techniques. Overall, the trend clearly shows that the proposed system provides more stable and accurate predictions, making it suitable for deployment in sensitive environments such as healthcare networks where minimizing false alerts is critical.

Chapter 6

Conclusions

In this project, a machine learning–based ransomware detection framework named RansomGuard ML was designed and implemented specifically for healthcare network environments. The increasing dependence of healthcare systems on digital infrastructure has made them highly vulnerable to ransomware attacks, which can disrupt critical services and compromise sensitive patient data. To address this issue, the proposed system focuses on behavioral analysis rather than traditional signature-based detection methods.

The framework utilizes system-level and network-level features such as file entropy, process activity, registry modifications, CPU usage, and network behavior to detect malicious activity. Machine learning models including Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN) were trained and evaluated using performance metrics such as accuracy, precision, recall, and false positive rate. Experimental results demonstrate that the Random Forest and the proposed optimized model achieve superior performance compared to other approaches, particularly in reducing false positives and improving detection reliability.

The comparative analysis with recent research further validates that the proposed system is competitive with existing methods while offering enhanced suitability for healthcare-specific environments. The inclusion of real-time monitoring capability and behavioral feature extraction makes the system more adaptive to evolving ransomware threats.

6.1 Work Carried Out

The work carried out in this project focuses on the design, development, and evaluation of RansomGuard ML, a machine learning–based ransomware detection framework specifically tailored for healthcare networks. The project was executed in a structured and systematic manner, covering all stages from problem understanding to system implementation and performance evaluation.

The first major step involved understanding the growing threat of ransomware attacks in healthcare environments. Hospitals and medical institutions are highly vulnerable due to their dependence on digital systems such as Electronic Health Records (EHR), network-connected medical devices, and real-time data access. Based on this understanding, the project aimed to develop a proactive detection system that could identify ransomware based on behavioral patterns rather than relying on traditional signature-based methods. To safely analyze ransomware behavior, a controlled virtual healthcare environment was created using virtualization tools such as VirtualBox and VMware. This environment simulated real-world hospital systems, including user workstations, file servers, and basic healthcare applications. The

use of virtualization ensured that ransomware samples could be executed without causing damage to actual systems or sensitive data. This sandboxed setup allowed for controlled experimentation and accurate observation of system behavior during both normal operations and malicious attacks.

Within this virtual environment, both benign applications and ransomware samples were executed. Benign applications included common software such as web browsers, document editors, and system utilities, which helped in establishing a baseline of normal system behavior. Ransomware samples were carefully selected and executed under monitored conditions to observe their impact on the system. During execution, detailed system-level and network-level data was collected. This included file operations such as creation, deletion, modification, and renaming; process activities such as process creation and termination; registry changes; CPU and memory usage; and network traffic patterns including inbound and outbound connections.

Once the data was collected, the next step involved preprocessing the raw data to ensure quality and consistency. The collected logs often contained noise, missing values, and redundant information, which could negatively impact model performance. Therefore, data cleaning techniques were applied to remove irrelevant or corrupted entries. Missing values were handled appropriately, and duplicate records were eliminated. The data was then structured into a tabular format such as CSV files, making it suitable for machine learning analysis. Normalization and feature scaling techniques were also applied to ensure that all features contributed equally during model training.

Feature engineering was a critical part of the project, as it directly influenced the effectiveness of the detection system. Relevant behavioral features were extracted from the preprocessed data to capture ransomware activity patterns. Key features included file entropy, which indicates the randomness of files and helps detect encryption; file modification rate, which identifies rapid changes to multiple files; process creation rate, which detects abnormal process behavior; and resource utilization metrics such as CPU and memory usage. Network-related features, including unusual outbound traffic and connections to unknown IP addresses, were also considered. These features were carefully selected to represent both system-level and network-level behavior.

After feature extraction, the dataset was labeled into two classes: benign and ransomware. This labeled dataset was then used to train supervised machine learning models. The project implemented three widely used algorithms: Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN). Each model was chosen for its ability to handle classification tasks and detect complex patterns in data. Random Forest was used for its robustness and ability to reduce overfitting, SVM for its effectiveness in high-dimensional spaces, and ANN for its capability to learn non-linear

relationships. The dataset was split into training and testing sets, typically using an 80:20 ratio. The training data was used to build the models, while the testing data was used to evaluate their performance on unseen samples. Hyperparameter tuning techniques such as grid search and cross-validation were applied to optimize model performance. The models were evaluated using standard performance metrics, including accuracy, precision, recall, and F1-score. Among the models, Random Forest showed the best performance, achieving high accuracy and maintaining a good balance between precision and recall, making it the most suitable choice for ransomware detection in healthcare environments.

In addition to model development, a real-time detection framework was designed to demonstrate practical implementation. The system was structured as a pipeline where incoming data is continuously monitored, processed, and analyzed using the trained model. Whenever suspicious behavior is detected, the system generates alerts to notify administrators. This real-time capability is essential in healthcare environments, where quick response to threats is critical to prevent data loss and system disruption.

To further demonstrate the usability of the system, a prototype full-stack application was developed using Node.js for the backend and React.js for the frontend. The backend handled data processing and detection logic, while the frontend provided a user-friendly interface for interaction. Users could input system parameters and receive instant feedback on whether the activity was safe or potentially malicious. A system logs module was also included to display detected anomalies, improving transparency and understanding of the system's operation. Throughout the project, several challenges were encountered, including handling noisy data, ensuring dataset balance, and safely executing ransomware samples. These challenges were addressed through careful preprocessing, controlled experimentation, and thoughtful system design. The project also emphasized minimizing false positives, as excessive alerts can disrupt healthcare operations.

Overall, the work carried out successfully demonstrates the application of machine learning and behavioral analysis for ransomware detection. The developed system provides an effective, scalable, and adaptable solution that can be further enhanced for real-world deployment. The project not only contributes to improving cybersecurity in healthcare but also provides valuable insights into the integration of machine learning techniques in practical security systems.

6.2 Key Achievements and Strengths

The RansomGuard ML project has achieved several significant milestones in the domain of cybersecurity, particularly in addressing the growing threat of ransomware attacks in healthcare networks. One of the most important achievements of this work is the successful design and development of a machine learning-based ransomware detection framework that focuses on behavioral

analysis rather than relying on traditional signature-based methods. This approach allows the system to identify unknown and zero-day ransomware variants, which are typically difficult to detect using conventional security mechanisms. By analyzing patterns of system activity instead of static signatures, the framework provides a more adaptive and future-ready solution for modern cybersecurity challenges. Another major achievement is the development of a controlled virtual healthcare environment for safe and effective data collection. This environment simulates real-world hospital systems, including file servers and user workstations, enabling the execution of both benign applications and ransomware samples without posing any risk to actual infrastructure. This setup ensured the collection of realistic and high-quality data, which is critical for training reliable machine learning models. The ability to safely study ransomware behavior in such an environment significantly strengthens the credibility and applicability of the proposed system.

The project also demonstrates a strong technical contribution through the implementation and comparative analysis of multiple machine learning algorithms, including Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN). Evaluating multiple models allowed for a deeper understanding of their strengths and limitations in ransomware detection. Among these, the Random Forest classifier emerged as the best-performing model, achieving higher accuracy and lower false positive rates. Its ensemble learning capability makes it particularly effective in handling complex and high-dimensional datasets, which is a key requirement in cybersecurity applications.

A key strength of the RansomGuard ML framework lies in its comprehensive feature engineering process. The system utilizes a wide range of behavioral features such as file entropy, file modification rate, process execution patterns, registry changes, CPU utilization, and network traffic behavior. These features are carefully selected to capture the core characteristics of ransomware activity. By focusing on behavior rather than static indicators, the system can detect malicious activity at an early stage, often before significant damage occurs. This early detection capability is especially critical in healthcare environments, where delays can directly impact patient care and safety.

Another important strength of the project is its emphasis on reducing false positives while maintaining high detection accuracy. In healthcare systems, unnecessary alerts can disrupt critical operations and reduce trust in security systems. The RansomGuard ML framework achieves a balanced performance by optimizing evaluation metrics such as precision and recall, ensuring that the system is both reliable and practical for real-world use.

The project also stands out due to its practical implementation through a prototype full-stack application. By using technologies such as Node.js for the backend and React.js for the frontend, the system provides

an interactive and user-friendly interface for monitoring and detection. This demonstrates how machine learning models can be integrated into real-time applications, bridging the gap between theoretical research and practical deployment. The inclusion of a dashboard and system logs enhances usability and helps users better understand system behavior and alerts. Furthermore, the modular and scalable architecture of the system is a key strength. Each component of the framework—data collection, preprocessing, feature engineering, model training, and detection—is designed independently, allowing for easy updates and improvements. This modular design also supports integration with external security platforms such as Security Information and Event Management (SIEM) systems, enabling centralized monitoring and efficient threat management. In addition, the project contributes to addressing a critical research gap by focusing specifically on healthcare networks, which are often underrepresented in cybersecurity research. By considering the unique requirements of healthcare environments, such as real-time performance, data sensitivity, and system reliability, the framework provides a more domain-specific and practical solution compared to generic ransomware detection systems. The RansomGuard ML project successfully combines machine learning, behavioral analysis, and system design to deliver a robust, scalable, and efficient ransomware detection solution. Its ability to detect advanced threats, minimize false alarms, and operate in real-time makes it a strong foundation for future research and real-world implementation in healthcare cybersecurity.

6.3 Reflections on the Development Process

The development of the RansomGuard ML framework was a valuable learning experience that provided both technical knowledge and practical insights into real-world cybersecurity challenges. Throughout the project, several important lessons were learned during the design, implementation, and evaluation phases. One of the key reflections is the importance of data quality in machine learning systems. Collecting realistic and well-labeled data in a controlled virtual environment was challenging but essential for building an effective detection model. Ensuring proper preprocessing and feature selection significantly improved model performance and reduced false positives. Another important learning was the complexity of working with behavioral analysis. Unlike signature-based detection, behavioral approaches require careful identification of meaningful features such as file entropy, process activity, and network behavior. Selecting the right features and avoiding irrelevant data played a crucial role in achieving accurate results. During model development, it was observed that different machine learning algorithms perform differently based on the dataset and features used. While Random Forest provided stable and high accuracy, models like SVM and ANN required more tuning and computational effort. This highlighted the importance of algorithm selection and hyperparameter optimization in machine

learning projects. The implementation phase also provided practical exposure to system integration and full-stack development. Developing a prototype using Node.js and React.js helped in understanding how machine learning models can be integrated into real-time applications. It also emphasized the importance of designing user-friendly interfaces for effective system interaction. Several challenges were encountered during the project, including handling noisy data, maintaining dataset balance, and ensuring safe execution of ransomware samples. These challenges were addressed through preprocessing techniques, controlled experimentation, and careful system design. The development process enhanced understanding of machine learning, cybersecurity, and system design. It also demonstrated the importance of combining theoretical knowledge with practical implementation to build efficient and reliable solutions for real-world problems, particularly in critical sectors such as healthcare. Another important reflection from the development process was the significance of designing a structured workflow and maintaining proper documentation at each stage of the project. Managing multiple components such as data collection, preprocessing, feature engineering, model training, and system integration required a clear and well-organized approach. Without proper planning, it would have been difficult to track progress, identify errors, and ensure consistency across different modules. This experience highlighted the importance of project management skills in technical research and development.

6.4 Limitations of the Current System

Despite the successful development of the RansomGuard ML framework, the project has certain limitations that should be considered for future improvements. One of the primary limitations is the use of a simulated virtual healthcare environment for data collection. Although it provides a safe and controlled setup, it may not fully capture the complexity and variability of real-world hospital networks. As a result, the model's performance in actual deployment scenarios may vary.

Another limitation is the size and diversity of the dataset. The dataset is generated from a limited number of ransomware samples and system executions, which may not represent all possible ransomware variants. This can affect the model's ability to generalize to new and highly sophisticated attacks. The current system also relies on selected behavioral features, which, although effective, may not capture every aspect of ransomware behavior. Some advanced ransomware variants may use stealth techniques that avoid triggering the selected features, potentially reducing detection accuracy.

A significant practical limitation is that the prototype implementation uses simplified rule-based logic in the frontend-backend integration instead of directly deploying the trained machine learning models. While this demonstrates system workflow, it does not fully reflect real-time machine learning-based

decision-making. Additionally, real-time performance and scalability have not been fully tested in large-scale healthcare environments. Processing high volumes of data across multiple devices may introduce latency and require further optimization. Another limitation is the lack of model explainability, especially for complex models like Artificial Neural Networks (ANN). In critical environments such as healthcare, understanding why a system flags an activity as malicious is important for trust and decision-making. Finally, integration with real-world security infrastructure, such as enterprise-level SIEM systems and endpoint protection platforms, has not been fully implemented and tested.

Another limitation of the current system is the potential imbalance between detection accuracy and computational efficiency. While machine learning models such as Random Forest and Artificial Neural Networks provide high accuracy, they may require significant computational resources for real-time processing, especially in large-scale healthcare environments. This can impact system performance and increase response time during high network activity.

Additionally, the system currently does not incorporate continuous learning or model updating mechanisms, meaning it may not automatically adapt to newly emerging ransomware variants without retraining. The absence of real-time threat intelligence integration further limits the system's ability to respond to evolving cyber threats dynamically. Addressing these limitations will be essential to enhance the system's adaptability, efficiency, and long-term reliability in real-world deployments. Another important limitation of the current system is the absence of user behavior analysis and contextual awareness in the detection process. The framework primarily focuses on system-level and network-level indicators, but it does not consider user activity patterns such as login behavior, access frequency, or unusual user actions. In real-world healthcare environments, insider threats or compromised user accounts can also contribute to ransomware attacks. Without incorporating user behavior analytics, the system may miss certain attack scenarios where malicious activities are carried out using legitimate credentials. Furthermore, the system currently lacks comprehensive validation using real-time streaming data, which limits its ability to fully demonstrate continuous monitoring capabilities under dynamic conditions. Incorporating user-centric features and real-time adaptive monitoring mechanisms would significantly improve the robustness and effectiveness of the system in practical deployments.

Chapter 7

Future Scope

The RansomGuard ML framework establishes a strong foundation for ransomware detection in healthcare networks by combining behavioral analysis with machine learning techniques. However, with the rapid evolution of cyber threats and the increasing complexity of healthcare infrastructures, there are several opportunities for future enhancements and research directions that can significantly improve the system's performance, scalability, and real-world applicability.

One of the most important areas for future work is the integration of fully trained machine learning models into the real-time detection system. In the current implementation, the prototype uses simplified rule-based logic to simulate detection behavior. While this approach is useful for demonstrating system workflow, it does not fully utilize the capabilities of machine learning. Future versions of the system can incorporate trained models such as Random Forest, Support Vector Machine (SVM), and Artificial Neural Networks (ANN) directly into the deployment pipeline. This will enable the system to make more accurate and intelligent predictions based on learned patterns rather than predefined thresholds. Additionally, deploying models using efficient frameworks and APIs will support seamless real-time processing and faster decision-making.

Another significant direction for improvement is the expansion of the dataset used for training and evaluation. The current dataset is generated within a controlled virtual environment and includes a limited number of ransomware samples. Although this ensures safety and controlled experimentation, it may not fully represent the diversity and unpredictability of real-world healthcare environments. Future research should focus on collecting larger and more diverse datasets, including real-world system logs, network traffic, and ransomware samples from different sources. Incorporating data from multiple healthcare systems will improve the model's ability to generalize and detect new and sophisticated ransomware variants, including zero-day attacks.

The adoption of advanced machine learning and deep learning techniques also presents a promising avenue for future research. While traditional models such as Random Forest and SVM provide strong baseline performance, more advanced approaches like Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Long Short-Term Memory (LSTM) networks can capture complex temporal and spatial patterns in system behavior. These models are particularly useful for analyzing sequential data, such as time-series logs and network traffic, where ransomware activities often exhibit specific patterns over time. Hybrid models that combine multiple algorithms or integrate both statistical and deep learning techniques can further enhance detection accuracy and robustness.

Scalability and real-time performance are also critical factors for deploying the RansomGuard ML framework in practical healthcare environments. Hospitals and healthcare institutions operate with large-scale networks that generate high volumes of data continuously. Therefore, future improvements should focus on optimizing the system for high-performance computing, reducing latency, and ensuring efficient data processing. Techniques such as distributed computing, cloud-based deployment, and edge computing can be explored to handle large-scale data streams effectively. These enhancements will allow the system to operate efficiently across multiple devices and locations without compromising performance. Another important area of future development is the implementation of explainable artificial intelligence (XAI). One of the limitations of advanced machine learning models, especially deep learning systems, is their lack of transparency. In critical sectors such as healthcare, it is essential for administrators to understand why a particular activity is flagged as malicious. Future research should focus on integrating explainability techniques that provide clear and interpretable insights into model decisions. Methods such as feature importance analysis, SHAP (SHapley Additive exPlanations), and LIME (Local Interpretable Model-agnostic Explanations) can be used to improve trust and accountability in the system. Integration with enterprise-level cybersecurity infrastructure is another key direction for future work. The RansomGuard ML framework can be enhanced by connecting it with Security Information and Event Management (SIEM) systems, Endpoint Detection and Response (EDR) solutions, and intrusion detection systems. Such integration will enable centralized monitoring, automated threat correlation, and efficient incident response. It will also allow the system to work as part of a larger security ecosystem, improving overall protection against cyber threats in healthcare environments.

Future versions of the system can also incorporate automated response mechanisms to move beyond detection and enable proactive defense. For example, when ransomware activity is detected, the system can automatically isolate the affected device, block suspicious processes, or restrict network access to prevent further spread. Implementing such response strategies will reduce the time between detection and mitigation, minimizing potential damage to healthcare systems and patient data.

Another promising direction is the inclusion of adaptive and continuous learning capabilities. Instead of relying on static models trained on fixed datasets, the system can be designed to update itself dynamically based on new data and emerging threats. Techniques such as online learning and incremental learning can help the model adapt to evolving ransomware behaviors without requiring complete retraining. This will ensure that the system remains effective over time and can handle new attack techniques. Finally, the RansomGuard ML framework can be extended to detect a broader range

of cybersecurity threats beyond ransomware. These include phishing attacks, malware infections, insider threats, and advanced persistent threats (APTs). By expanding its scope, the system can evolve into a comprehensive cybersecurity solution for healthcare organizations, providing multi-layered protection against various types of attacks. In conclusion, while the current RansomGuard ML framework demonstrates strong potential for ransomware detection, future enhancements focusing on real-time deployment, advanced models, scalability, explainability, and integration will significantly improve its effectiveness. These developments will contribute to building a more secure, intelligent, and reliable cybersecurity system for modern healthcare infrastructures.

Area	Current Limitation	Proposed Future Improvement	Expected Benefit
Model Deployment	Uses rule-based prototype	Integrate trained ML models (RF, SVM, ANN)	Higher accuracy and real-time intelligent detection
Dataset	Limited and simulated data	Use large-scale real-world healthcare datasets	Better generalization and robustness
Detection Techniques	Traditional ML models	Apply deep learning (CNN, RNN, LSTM)	Improved detection of complex patterns
Scalability	Not tested in large environments	Cloud-based and distributed deployment	Efficient handling of large data volumes
Real-Time Performance	Limited optimization	Low-latency processing and stream analysis	Faster detection and response
Explainability	Black-box models	Use XAI techniques (SHAP, LIME)	Improved transparency and trust
System Integration	Limited integration	Connect with SIEM and EDR systems	Centralized monitoring and response
Automated Response	Detection only	Add auto-isolation and threat blocking	Faster mitigation of attacks
Adaptability	Static models	Implement online/incremental learning	Continuous improvement and adaptability
Scope	Focus on ransomware	Extend to phishing, malware, insider threats	Comprehensive cybersecurity solution

Table 5.1: Future Enhancements and Research Directions

References

- [1] R. Mahajan, “Phishing website detection using machine learning algorithms,” ResearchGate, 2016.
- [2] A. K. Jain and B. B. Gupta, “Towards detection of phishing websites on client-side using machine learning based approach,” Springer, 2017.
- [3] C. Liu and B. Lang, “Finding effective classifier for malicious URL detection,” in Proc. ACM, 2018.
- [4] O. K. Sahingoz, E. Buber, O. Demir, and B. Diri, “Machine learning based phishing detection from URLs,” Expert Systems with Applications, vol. 117, pp. 345–357, 2019.
- [5] Scikit-learn, “Support Vector Machines (SVM) documentation,” 2023. [Online]. Available: <http://scikit-learn.org/stable/modules/svm.html>
- [6] University of New Brunswick CIC, “UNB URL dataset,” 2016. [Online]. Available: <https://www.unb.ca/cic/datasets/url-2016.html>
- [7] A. Abunadi, A. Zainal, and O. Akanbi, “Feature extraction process: A phishing detection approach,” in Proc. IEEE, 2013.
- [8] Y. Zhang, J. Hong, and L. Cranor, “CANTINA: A content-based approach to detecting phishing web sites,” in Proc. 16th Int. Conf. World Wide Web (WWW), 2007, pp. 639–648.
- [9] A. Bergholz et al., “New filtering approaches for phishing email,” Journal of Computer Security, vol. 18, no. 1, pp. 7–35, 2010.
- [10] M. Khonji, Y. Iraqi, and A. Jones, “Phishing detection: A literature survey,” IEEE Communications Surveys & Tutorials, vol. 15, no. 4, pp. 2091–2121, 2013.
- [11] A. K. Jain and B. B. Gupta, “Phishing detection: Analysis of visual similarity-based approaches,” Security and Privacy, vol. 1, no. 1, p. e9, 2016.

- [12] R. Basnet, A. H. Sung, and Q. Liu, "Rule-based phishing attack detection," *Computers & Security*, vol. 39, pp. 17–30, 2012.
- [13] R. Verma and N. Hossain, "Semantic feature selection for text with application to phishing email detection," in *Proc. ACM CODASPY*, 2014, pp. 47–58.
- [14] R. M. Mohammad, F. Thabtah, and L. McCluskey, "Intelligent phishing detection system using association rule mining," *Journal of Network and Computer Applications*, vol. 36, no. 1, pp. 324–333, 2014.
- [15] J. Ma, L. K. Saul, S. Savage, and G. M. Voelker, "Beyond blacklists: Learning to detect malicious web sites from suspicious URLs," in *Proc. 15th ACM SIGKDD*, 2009, pp. 1295–1304.
- [16] G. Xiang, J. Hong, C. P. Rose, and L. Cranor, "Cantina+: A feature-rich machine learning framework for detecting phishing web sites," *ACM Transactions on Information and System Security*, vol. 14, no. 2, pp. 1–28, 2011.
- [17] N. Abdelhamid, A. Ayesb, and F. Thabtah, "Phishing detection based associative classification data mining," *Expert Systems with Applications*, vol. 41, no. 13, pp. 5948–5959, 2014.
- [18] R. Noor and N. Salim, "Phishing website detection using ensemble learning," *Journal of Information Security and Applications*, vol. 35, pp. 100–112, 2017.
- [19] I. Abdullahi and A. Salihu, "URL-based phishing detection using machine learning: A comprehensive survey," *Journal of Cybersecurity Research*, vol. 3, no. 2, pp. 45–68, 2020.
- [20] J. Ma, L. K. Saul, S. Savage, and G. M. Voelker, "Identifying suspicious URLs: An application of machine learning," *IEEE Security & Privacy*, vol. 9, no. 6, pp. 12–19, 2011.

APPENDIX Sample Dataset

Below is a sample of the dataset used in the RansomGuard ML framework. The dataset contains system-level and network-level behavioral features collected from the virtual healthcare environment. Each instance is labeled as either Benign or Ransomware.

File Entropy	File Modification Rate	CPU Usage (%)	Network Activity	Label
7.8	65	85	72	Ransomware
6.2	10	30	20	Benign
7.5	58	80	75	Ransomware
5.9	12	25	18	Benign
8.1	70	90	80	Ransomware
6.0	15	35	22	Benign
7.9	60	88	78	Ransomware